

SONIC: Symmetry-Oriented Non-redundant Internal Coordinates

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Abstract

Internal coordinates are the natural language of molecular structure, yet automatic constructions either retain redundancy or produce delocalized, weight-dependent bases and remain difficult to generalize to fused and bridged rings, high coordination, exact and local symmetry, and weakly bound fragments. We introduce a deterministic, domain-first protocol that accepts a frozen molecular state or provider-neutral Cartesian, topology, or primitive-coordinate input. After constructing all stretches, it identifies ring and local XY_3/XY_2 domains, assigns their primitive ownership, builds their complete special-coordinate blocks, and only then constructs ordinary coordinates from the unassigned residual space. Equivalent XY_3 and acyclic XY_2 termini receive normalized collective rotor coordinates, whereas branch deformations are represented by orthonormal local bending and out-of-plane combinations; ambiguous or heterosubstituted motifs revert deterministically to an ordinary one-dihedral coordinate. Validation across acyclic, cyclic, fused, bridged, high-symmetry, high-coordination, non-covalent, and metal–ligand systems gives complete target rank, analytic-versus-finite-difference first-derivative agreement, rotation/permutation covariance, Python–Fortran agreement, stable local symmetry-adapted combinations, and reproducible external serialization;

Gaussian 16 provides an independent parser and solver for one external representation of the generated coordinates. The resulting frozen contract retains chemically readable decompositions, symmetry, rank diagnostics, analytic first derivatives, and provenance, thereby providing a constructive counterpart to natural internal coordinates while leaving optimization, finite internal-to-Cartesian realization, force fields, and higher derivatives to downstream consumers.

1 Introduction

The Wilson internal-coordinate formalism gives a compact relation between Cartesian displacements $\Delta\mathbf{x}$ and internal-coordinate displacements $\Delta\mathbf{q}$,

$$\Delta\mathbf{q} = \mathbf{B}\Delta\mathbf{x}, \tag{1}$$

and remains the standard framework for vibrational analysis, potential-energy distributions and internal-coordinate force constants.¹ The formalism itself does not solve the constructive problem: one must still choose which coordinates define the molecular model. That choice is benign in small hand-built $\mathbf{Z}$ matrices, but it becomes a central scientific issue when the same coordinate model is expected to support symmetry-constrained optimization, semi-experimental structural refinement, ring puckering, weak complexes and metal or high-coordination environments.

Symmetry-Oriented Non-redundant Internal Coordinates (SONIC) are designed to make this choice explicit and reproducible. Table 1 defines the permanent boundary used below. The essential separation is that the Symmetry-Mapped Internal-coordinate Template Handler (SMITH) is the standalone tool that constructs SONIC contracts, while SONIC is the coordinate family: the non-redundant generalized internal-coordinate (GIC) basis, analytic Wilson matrix and symmetry labels. Applications choose how those coordinates are displaced, optimized, fitted or differentiated beyond first order.

Table 1: Permanent boundary of the standalone SMITH/SONIC work.

Name	Role
Input provider	Supplies Cartesian geometry and, optionally, topology, redundant primitives, symmetry operations, fragments, interaction centers and continuous chemical descriptors. For plain Cartesian input, standalone SMITH embeds a revision-pinned perception subset for ordinary topology, point-group operations, atom permutations and primitive coordinates.
SMITH– SONIC	Coordinate-construction tool and its coordinate contract. SMITH accepts a complete frozen state, a supplied topology, a supplied redundant primitive set, or plain Cartesian input through its bundled perception frontend. It performs SONIC construction, special/composite coordinate generation, symmetry adaptation, analytic $\mathbf{B}$ evaluation, human-readable reporting, Gaussian-16-compatible export and provenance.
Application layer	Consumes the frozen contract. Internal-to-Cartesian realization, optimization, scanning, force-field construction, property models, Hessian transport and higher derivatives are deliberately outside SMITH.

Within the integrated MATRIX implementation these generic roles have one explicit mapping, stated here once to avoid coupling the standalone method to the suite architecture. ORACLE supplies symmetry, atom permutations, redundant primitives and their Wilson matrix; SMITH consumes that frozen state and constructs SONIC without re-perception; LINK owns finite internal-to-Cartesian realization for optimization, scans and externally proposed points; ARCHITECT constructs $\mathbf{B}'$ on demand for nonstationary Hessian transformations; MORPHEUS delegates any coordinate realization to LINK; and SENTINEL exchanges SONIC points with LINK without consuming $\mathbf{B}$.

The standalone executable preserves this scientific boundary by copying only the required perception implementation into its release snapshot. It can therefore infer symmetry from an XYZ file without installing ORACLE, but the algorithm, thresholds and schema remain those of the pinned ORACLE revision. If a complete frozen state or an explicit symmetry section is supplied, that record is authoritative and the embedded kernel does not replace it. Figure 1

summarizes this construction boundary and its downstream applications. In the figure, GF/PED denotes the Wilson GF matrix method and potential-energy-distribution analysis, DVR denotes discrete-variable representation and PES denotes potential-energy-surface fitting.

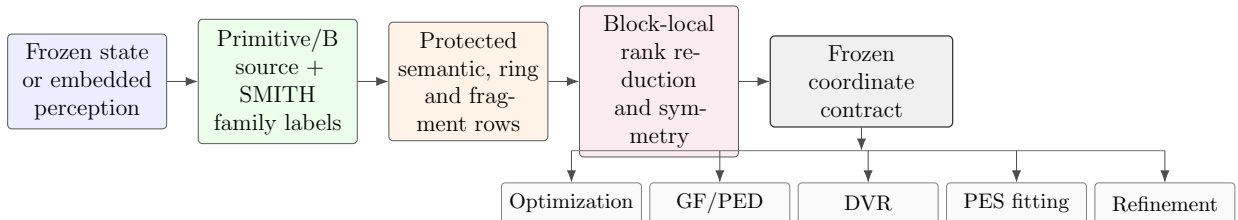


Figure 1: SMITH/SONIC construction boundary. Cartesian geometry and optional topology or redundant primitive/Wilson-row data lead to a frozen coordinate contract; downstream algorithms consume but do not modify that object.

The frozen coordinate contract is directly inspectable rather than an opaque numerical basis. For every accepted SONIC, SMITH retains the ordered primitive sources, the corresponding coefficient row, family and symmetry labels, unit, protection status, analytic Wilson row, rank diagnostics and provenance. It also writes a human-readable expansion and an optional stored $-h, 0, +h$ Cartesian trajectory, so the algebraic definition and the molecular motion can be checked independently (Figure 2).

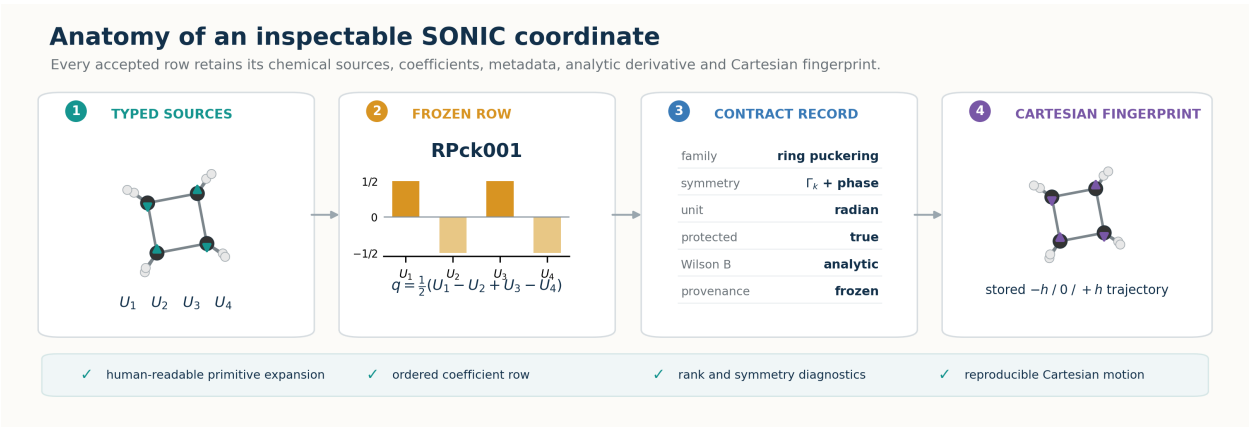


Figure 2: Inspectable SONIC ring-puckering record. Typed sources and stored coefficients define a named coordinate; metadata, analytic Wilson row, provenance and an optional $-h, 0, +h$ Cartesian fingerprint audit it. Arrows are enlarged and do not prescribe a finite back-transformation.

The central novelty is a deterministic domain-first construction, not the introduction of another undifferentiated list of internal-coordinate formulas. A useful generator must choose a non-redundant basis while satisfying four requirements at once: correct vibrational rank, chemical locality, symmetry adaptation without cross-family mixing, and protection of special coordinates. SMITH now makes the last requirement constructive: after the bonded stretching frame is fixed, rings and eligible XY_3/XY_2 centers are recognized as coordinate domains; each domain builds its complete natural block and claims the primitives it owns before any ordinary bend, out-of-plane row or torsion is admitted. Ring puckering, collective local rotors, ring-center distances, fragment translations and fragment orientations therefore cannot be silently replaced by an algebraically equivalent ordinary coordinate during rank reduction. The result is not merely a coordinate list but a persistent object that downstream modules can audit, reuse and compare against independent active-space constructions.

Equally important, SONIC is not a new coordinate language. Modern GIC implementations already provide expressive syntax for nonlinear and mixed coordinate expressions. The novelty here is the preceding step: a deterministic generator that converts a frozen molecular state into a rank-complete, symmetry-consistent coordinate basis that can be serialized as Gaussian-16-readable GICs or consumed directly by downstream modules. Such a SONIC serialization is only one of the many possible uses of the general GIC language. The standalone command-line tool is named SMITH (Symmetry-Mapped Internal-coordinate Template Handler). Symmetry is used in the classical molecular sense: point-group operations map coordinate rows into the same space with the corresponding atom permutation and vector or pseudovector signs. Orthogonality enters through normalized Wilson-row rank selection and orthonormal symmetry-adapted linear-combination (SALC) coefficient vectors inside homogeneous blocks, not through global orthogonality in the kinetic $\mathbf{G}$ metric.

2 Relationship to Existing Coordinate Models

Table 2 summarizes the position of SONIC relative to the main coordinate strategies used in quantum chemistry and molecular spectroscopy. The comparison is deliberately constructive. SONIC is not a replacement for the Wilson formalism, natural internal coordinates or translation-rotation coordinates; it is a general layer that decides how such ingredients are assembled into one frozen, reusable coordinate object.

Table 2: How SONIC is positioned relative to established coordinate strategies.

Strategy	Strength	Relationship to SONIC
Z matrices	Compact, local and interpretable for small molecules.	Tree dependence, singularities, and manual choices make them fragile for rings, symmetry and automated workflows. ²
Redundant internal coordinates	Robust for optimization and easy to generate from a graph. ³⁻⁵	They do not define a unique active basis for vibrational analysis, symmetry projection or least-squares refinement.
Delocalized internal coordinates	Automatic non-redundant bases can be obtained from global rank decompositions. ⁶	The coordinates can mix stretches, bends, torsions and weak contacts; weights used to control this mixing are method choices rather than chemical observables.
Natural internal coordinates	Highly interpretable symmetry and force-field variables for well understood local motifs. ⁷⁻⁹	General construction is difficult for fused or bridged rings, high coordination, fragments, interaction centers and arbitrary point groups.
Translation-rotation internal-coordinate (TRIC)/geomeTRIC coordinates	Explicitly represents collective fragment translations and rotations, with the rotations described by an exponential map of quaternions. ¹⁰	SONIC adopts this exponential-map parametrization, but not the complete TRIC construction: it defines the rigid-body directions relative to a reference fragment and retains them as a protected, typed and symmetry-labelled source family; alternatively, the intermolecular model may be constructed from typed pseudo-bonds and pseudo-cycles.
Gaussian 16 GIC syntax	Very general symbolic coordinate expressions, including nontraditional geometric parameters, sparse derivative rows and automatic symbolic differentiation, can be evaluated by the optimizer. The recent work substantially extends this general framework. ¹¹	SONIC supplies one particular, automatically constructed family of definitions that can be passed to GIC syntax; the present tests use only features available in the commercial Gaussian 16 Rev. C.01 installation, not the development-version extensions reported in the recent work.
SONIC	Builds a rank-complete, type-labelled, symmetry-adapted coordinate basis from a validated molecular state.	Weak complexes can be represented either by fragment/center coordinates or by typed pseudo-bonds and pseudo-cycles; the remaining development frontier is the breadth of the pseudo-cycle validation corpus.

This positioning is important for the narrative of the method. Primitive redundant internals and delocalized internals are excellent tools for geometry optimization, where the main criterion is stable movement on a potential-energy surface. A semi-experimental fit or

a Wilson GF analysis imposes stricter requirements: the fitted or analyzed variables must be identifiable, symmetry consistent and auditable as the same objects across independent modules. Natural internal coordinates meet these requirements when the local chemistry is known in advance. Bércecs extended their use to transition-metal skeletal degrees of freedom,⁷ von Arnim and Ahlrichs generalized the automatic construction while still identifying multiply fused ring systems as a hard case,⁸ and Swart and Bickelhaupt emphasized the need to distinguish strong and weak coordinates in optimization.⁹ SONIC turns this lineage into a frozen construction for spectroscopy and refinement by combining chemically typed primitive generation, block-local rank reduction, special-coordinate protection and point-group projectors.

The recent Gaussian 16 GIC paper provides the most direct complementary framework for external serialization. Its Supporting Information demonstrates the broad scope of the GIC language with nonlinear water coordinates, formamide proton-transfer sum and difference variables, methane and ammonia XY_3 -type symmetrized combinations, polar and elliptic Li_2 -Ar scan variables, fragment rotor coordinates for benzene dimers and water clusters, and quaternion fragment coordinates for ferrocene.¹¹ These new capabilities are particularly interesting and provide natural targets for future interoperability tests. They were not used in the present work: the cited study states that its calculations employed a development version of the Gaussian suite, whereas all external tests reported here were restricted to the functionality available in the commercial Gaussian 16 Rev. C.01 installation. Accordingly, the examples below are motif-level construction tests or Rev. C.01 serialization tests, not validations of the new development-version implementation. These examples illustrate the generality of an expressive coordinate language. Within that broader setting, Gaussian 16 can evaluate and differentiate the SONIC expressions once they have been supplied, just as it can treat many other GIC definitions. SONIC provides an optional automatic route to one such set of objects when they follow from topology, local equivalence, point-group symmetry, fragments, ring centers, bond centers or semantic coordinate requests.

This complementarity is central to the positioning of the method. The Gaussian 16 GIC facility is a general coordinate-description language in which a user can define primitive, nonlinear, redundant or mixed expressions. SONIC constructs one possible coordinate model before an optimizer, force-field fit or refinement uses it, and records why each special or ordinary coordinate was accepted. GIC syntax is therefore one supported serialization route for SONIC; SONIC neither restricts the scope of that syntax nor replaces the other ways in which it can be used.

For repeated automatic workflows, molecule-specific GIC definitions for local XY_n -type combinations, proton-transfer sums and differences, fragment translations or ring-puckering components can instead be generated from the frozen molecular state. The distinction is especially clear for XY_3 . The Gaussian construction spans two differential branch torsions and their common rotation.¹¹ SMITH uses the same physical local space but chooses a non-redundant, symmetry-transparent gauge: branch deformations are carried by orthonormal local bending SALCs, while the torsional block retains the normalized common rotor averaged over every admissible skeleton anchor. The resulting coordinate is independent of an arbitrary choice of one hydrogen or one anchor and has the correct threefold periodicity for equivalent terminal branches. The analogous XY_2 construction is restricted to acyclic centers, uses a twofold collective rotor, and assigns the complementary in-plane and out-of-plane motions to the same owned domain. When equivalence or a unique skeleton branch is not established, SMITH does not impose a local symmetry: it records the failed recognition and uses the deterministic one-dihedral rule. Thus variables that follow from the frozen molecular state are generated and protected automatically; an explicit user request is instead interned as a semantic coordinate rather than retained as an opaque handwritten expression.

3 Molecular State and Atom Equivalence

SONIC does not rediscover the molecule at every downstream step. Its formal input boundary is provider-neutral: Cartesian geometry plus either a frozen topology or an ordered redundant primitive set with the information needed to reconstruct its Wilson B rows. A complete molecular state may additionally contain rings, point-group operations, fragments, interaction centers and continuous chemical-perception descriptors. If those data are already present in the `xyzin` file, SONIC consumes them as part of the frozen state. The provider may be the bundled perception frontend, another program or a hand-audited input; its identity is not part of the SONIC theory. The bundled frontend includes the revision-pinned symmetry, topology and ordinary primitive-internal-coordinate (PIC) subset required for standalone construction. A rich state can define the molecular graph and cycle basis, point-group operations and atom permutations, atom equivalence, effective atomic numbers, continuous charge, covalency, delocalization and strain descriptors, bond-order components and synthon indices. SMITH consumes these records when present. The complete state may also supply ordinary stretch, bend, linear-bend, dihedral and out-of-plane source definitions and their reference B matrix. Before accepting those ordinary angular and torsional candidates, SMITH constructs a frozen domain-ownership map. Rings, butterfly units, eligible XY_3 and acyclic XY_2 centers, fragments, pseudo-bonds, interaction centers and protected semantic requests claim their source primitives in that map. SMITH then adds their complete special blocks and constructs the ordinary residual chart only from primitives not already owned by a special domain. This ordering is part of the coordinate contract, not an implementation detail. The output is a frozen #GIC definition: consumed and augmented primitive coordinate records, final generalized coordinates, coefficients, family labels, irreducible representations, total-symmetric subsets and the metadata required to reconstruct analytic **B** rows. This is the end of the SMITH contract.

For standalone use the required boundary can be supplied in one extended-XYZ file conforming to `matrix.smith.input.v1`. Four provider-neutral profiles are accepted: a

complete frozen state, supplied topology, supplied primitives or a minimal Cartesian geometry processed by the bundled perception frontend. Their exact syntax, validation rules and executable examples are given in the Supporting Information (SI), Sec. S1. Thus SMITH is operationally standalone while preserving a precise boundary between perception and coordinate construction: the embedded subset provides the frozen input state, and SMITH proper constructs the SONIC chart from that state.

The richer optional descriptors follow continuous molecular perception introduced in Proxima,¹² and molecular/spectral synthons developed for accurate structural templating.^{13–15} Initial local equivalence is therefore not based only on element symbols. The input state may supply continuous atom-perception and synthon descriptors, and SONIC uses them to make reproducible local choices. The present construction uses effective atomic numbers that combine nuclear charge with local coordination, distance-derived bond-order information, angular rigidity and delocalization-like terms. These continuous descriptors order atoms and ligands in cases where a pure graph label would be ambiguous. The topology layer also exposes synthon descriptors—charge, covalency, delocalization, strain, effective nuclear charge and incident $\sigma/\pi/\pi\pi$ indices—which provide a natural route to more refined equivalence classes. In the present contract, this continuous perception enters ring orientation, ligand grouping, local stretch equivalence, high-coordination templates and the one-dihedral exocyclic torsion rule. In particular, ring puckering diagnostics report the first- and second- π indices of every endocyclic central bond alongside its topological order and flexibility factor. Triple-bond character is therefore not collapsed into the same aggregate $\max(\text{BO} - 1, 0)$, where BO denotes bond order, used historically for all multiple bonds. The reproducibility statements below are conditional on these descriptors being part of the frozen input state. In other words, SONIC does not assume an unstated human perception step: the descriptor values, perception version, topology and symmetry operations are stored as contract provenance, and a change in any of them is a different molecular-state stratum.

4 Coordinate Functions and Downstream Models

The frozen SONIC basis is the first-order coordinate frame. Many downstream models, however, are better written in functions of those coordinates than in the raw coordinates themselves. This is especially true for force fields and PES fitting, where the physically smooth variable may be an inverse distance, an exponential contact coordinate, a trigonometric torsional term or a ring-puckering descriptor. We therefore distinguish base coordinates $\mathbf{q}$, which define rank and symmetry, from model coordinates $\mathbf{y} = f(\mathbf{q})$, which are evaluated by downstream modules.

For any differentiable coordinate function $y_a = f_a(\mathbf{q})$, the Wilson row used by a fit or force-field model follows from the chain rule,

$$\frac{\partial y_a}{\partial \mathbf{x}} = \sum_i \frac{\partial f_a}{\partial q_i} \frac{\partial q_i}{\partial \mathbf{x}}. \quad (2)$$

Thus the same frozen $\mathbf{B}$ service can support nonlinear coordinate functions without changing topology, rank selection or symmetry labels. Examples include r^{-1} , $\exp(-\alpha r)$ and damped contact functions for distances; $\sin \theta$, $\cos \theta$, $\sin n\phi$ and $\cos n\phi$ for bends and torsions; and Cremer–Pople-like q and ϕ descriptors derived from linear ring-puckering (*RPck*) components. The same layer also covers the practical variables used in the Gaussian GIC examples: bond-sum and bond-difference coordinates for proton transfer, symmetrized XY_n local combinations, polar or elliptic functions of intermolecular Cartesian components, and quaternion or exponential-map functions of fragment orientation.¹¹ In a GIC input language these are written as explicit expressions or requested through a specific option; in SONIC they are downstream functions of generated base coordinates or protected fragment coordinates, with the chain rule preserving the connection to the frozen Wilson rows. Recent work by Lema-Saavedra and Fernández-Ramos shows why this separation is useful for rings: Cremer–Pople amplitudes provide an efficient conformational search space, while chemically viable 3D geometries require ring closure, angular redistribution, treatment of rigid or multiple bonds

and invariant local references for substituents.¹⁶ The active Gaussian-readable coordinate basis should remain linear and rank complete, while such nonlinear functions become named observables or model variables in the Wilson *GF* matrix method, potential-energy-distribution (PED) analysis, DVR, PES-fitting or refinement workflows.

Conformational reconstruction from puckering variables is therefore not external to this workflow. In the DVR layer, a one- or multidimensional grid in *RPck*-, *Q*- or ϕ -like variables must be associated with physically closed ring geometries before electronic-structure points, metric terms or wavefunctions can be interpreted. The reconstruction layer of Ref. 16 provides precisely this coordinate-to-geometry step: closure constraints, angular redistribution, rigid-bond treatment and substituent reference frames belong above the frozen SONIC basis, but they are natural consumers of it. A DVR treatment can then recompute the Wilson metric on the reconstructed structures, or use an explicitly validated reduced metric, without introducing a second private ring-coordinate definition.

This distinction also prevents a common ambiguity in the word “GIC.” A general GIC expression can be an optimizer coordinate. A SONIC coordinate definition is reusable: it specifies the base coordinate space, the analytic derivatives and the symmetry transformation rules from which later functional variables can be built. Implementing these coordinate functions systematically is part of the next development layer, but it does not require a different coordinate-generation theory.

SONIC therefore exposes a small semantic coordinate layer rather than a second GIC language. PROTECT requests name base objects such as distances, torsions, ring puckering, butterfly coordinates, fragment translations or center distances. Independent requests enter rank and symmetry selection with the same priority as automatically generated protected coordinates. OBSERVABLE requests define functions such as inverse distances, trigonometric terms, linear combinations, Cremer–Pople variables or proton-transfer coordinates; they are evaluated later as $y = f(\mathbf{q})$ by Eq. (2). Duplicate user rows are rejected, row-equivalent user rows substitute the automatic primitive with provenance recorded, and partial symmetry

orbits are diagnosed before local SALC fallback.

5 The Frozen Coordinate Contract

The central output of SONIC is a persistent coordinate object, not a transient list of internal coordinates. This distinction is important because the same coordinate object must be used by independent modules, with no private reconstruction of topology, symmetry or rank choices.

Definition 1. A frozen coordinate contract is the tuple

$$\mathcal{C} = (\mathcal{P}, \mathcal{G}, \mathbf{B}, \Sigma, \mathcal{M}).$$

Here $\mathcal{P}_0 = (p_1, \dots, p_{m_0})$ is the ordered redundant primitive/B source supplied at the input boundary and $\mathcal{P} \supseteq \mathcal{P}_0$ is the typed source space after any SMITH special/composite augmentation; $\mathcal{G} = (\mathbf{g}, \mathbf{C}_\mathcal{G})$ is the accepted non-redundant generalized-coordinate basis together with its coefficient matrix; $\mathbf{B}$ contains the analytic Wilson rows of the accepted basis; Σ contains point-group transformation matrices, irreducible-representation labels and SALC coefficients; and $\mathcal{M}$ contains the frozen molecular state, source-family labels, protection flags, family priority order, numerical tolerances, semantic grammar and contract schema versions, diagnostics and provenance. Equivalently, with $\mathbf{p} = (p_1, \dots, p_m)^T$,

$$\mathbf{g} = \mathbf{C}_\mathcal{G} \mathbf{p}, \quad g_a = \sum_{j=1}^m c_{aj} p_j, \quad \mathbf{B} = \frac{\partial \mathbf{g}}{\partial \mathbf{x}} = \mathbf{C}_\mathcal{G} \mathbf{B}(\mathcal{P}).$$

The tuple is the object consumed by GF/PED, DVR, PES-fitting and refinement workflows. It is also the object that makes the method auditable: a downstream result can be traced to the primitive candidates that were generated, the rows that were accepted or rejected, the rank tolerance that was used and the symmetry block in which each final coordinate was projected.

Let $X = \mathbb{R}^{3N}$ be the Cartesian displacement space and let T denote the rigid-body tangent space generated by overall translations and rotations. For a nonlinear molecule, $\dim T = 6$; for a linear molecule, $\dim T = 5$. Scalar intramolecular coordinates have first-order rows in the annihilator

$$X_{\text{int}}^* = \{\ell \in X^* : \ell(t) = 0 \text{ for all } t \in T\},$$

whose target dimension is the vibrational rank $r = 3N - \dim T$. Vector-like fragment translations and orientations are stored as relative or body-fixed generalized coordinates and are audited against the same external-mode-free active space. Each primitive candidate $p_i \in \mathcal{P}$ contributes an analytic row $\mathbf{b}_i = \partial p_i / \partial \mathbf{x}$, and

$$\mathbf{B}(\mathcal{P}) = \begin{bmatrix} \mathbf{b}_1 \\ \vdots \\ \mathbf{b}_m \end{bmatrix}, \quad \mathcal{R}(\mathcal{P}) = \text{span}\{\mathbf{b}_1, \dots, \mathbf{b}_m\}.$$

The constructive goal is to choose an ordered subset $\mathcal{A} \subseteq \mathcal{P}$ such that

$$\text{span } \mathbf{B}(\mathcal{A}) = \mathcal{R}(\mathcal{P}), \quad |\mathcal{A}| = \text{rank } \mathbf{B}(\mathcal{P}) = r,$$

when the primitive candidate set spans the full internal row space. If this condition fails, the build is not silently repaired; the contract records the rank defect or excess and the molecular state must be reconsidered.

The novelty of SONIC lies in the constraints imposed on this basis selection. The accepted set $\mathcal{A}$ is not just any basis of $\mathcal{R}(\mathcal{P})$. It is a chemically ordered basis selected under a source-family preorder: special fragment, center and ring coordinates are tested before ordinary rank-completion candidates, and typed chemical families are reduced before unrelated families are allowed to compete. This replaces the search for an arbitrary non-redundant basis by the construction of a basis with explicit semantic priorities.

6 Construction Principles

The construction is easiest to state as a sequence of constraints on the candidate space.

1. All covalent bonds in the validated topology first receive canonical stretch rows. SMITH generates them in minimal mode or validates and classifies supplied candidates. Their persistent ordering is a graph property, not a geometry-dependent consequence of a local SVD.
2. The complete special-domain inventory is then frozen before ordinary angular or torsional generation. It includes elementary and fused ring domains, butterfly units, eligible XY_3 centers, eligible acyclic XY_2 centers, fragments, pseudo-cycles and explicit semantic or interaction-center requests. A center with fully equivalent ligands is not forced into an $XY_n - Z$ pattern: Z must be a distinct, unique branch.
3. Every recognized special domain constructs its full coordinate block and registers the primitive motions it owns. Ring domains generate deformation and puckering components; XY_3 domains generate five orthonormal bending SALCs and one collective rotor where an admissible anchor exists; XY_2 domains generate two in-plane combinations, one oriented out-of-plane coordinate and one collective rotor. Fused and condensed rings retain their shared-edge and butterfly semantics.
4. Ordinary local valence angles, linear bends, out-of-plane rows and exocyclic torsions are generated only from the residual, unowned primitive space. Consequently, a standard row cannot duplicate or displace a special motion. Non-ring bonds outside a recognized collective-rotor domain use exactly one deterministic dihedral selected from the admissible substituent pairs; ONEDIH does not denote an averaged torsional orbit.
5. Optional local-pseudosymmetry SALCs and high-coordination templates are then formed within the remaining chemically homogeneous center blocks. Named templates

are optional: every coordination number above four has a template-independent ligand-equivalence fallback.

6. The final non-redundant basis is selected by analytic **B**-row rank, using the fixed ownership and family order. Exact molecular symmetry is applied only after a valid non-redundant basis exists and only inside typed source blocks.

This ordering differs from a global diagonalization of the full primitive metric, whether expressed as $\mathbf{B}\mathbf{B}^T$, as a row-space projector such as $\mathbf{B}\mathbf{B}^+$, or as a single singular value decomposition (SVD) of every redundant primitive at once. Such decompositions are mathematically useful, but their eigenvectors depend on how the primitive rows have been scaled. A stretch measured in angstrom, a bend measured in radians, a torsion, a fragment translation and a center distance have different natural units and different chemical roles. Changing their weights can change which delocalized coordinates are retained. SONIC instead uses rank-revealing linear algebra inside chemically typed blocks and reserves the final cross-block competition for a fixed analytic **B**-rank test. This does not make the construction unique in an abstract mathematical sense. It replaces hidden numerical weighting choices by an explicit chemical priority order whose provenance is stored in the frozen definition.

7 Primitive Source and SMITH Coordinate Families

The primitive source layer is intentionally redundant. Redundancy is useful at this stage because it allows chemical rules to create all plausible local deformations before the rank selector removes dependent rows. The important restriction is that source primitives receive SMITH family labels that survive reduction, symmetry adaptation and output.

7.1 Bonded Coordinates

Every covalent edge supplies one stretch. Equivalent stretches can be converted to local sums and differences using a block-local SVD of their analytic **B** rows, but the source block

remains a stretch block. Ordinary bends are generated from pairs of neighbors around each center. Nearly linear centers produce two linear-bend components, because one ordinary angular derivative becomes singular at linearity.

For three- and four-coordinate centers, SONIC uses local equivalence classes to construct chemically recognizable bend combinations: symmetric deformations, rocking coordinates and tetrahedral or WXY_3/W_2XY_2 -like patterns. Recognition is conservative and persistent. Terminal identical atoms are exactly permutable even when a distorted geometry gives slightly different continuous descriptors; nonterminal branches must also have the same rooted graph environment after removal of the center. Conversely, atomic number and coordination number alone are insufficient to merge branches. An XY_2 center is excluded whenever the center belongs to a cycle, and both XY_2 and XY_3 require a unique Z branch distinct from the equivalent Y class. These rules prevent atom numbering or accidental geometric pseudosymmetry from creating a special domain. For coordination numbers between five and nine, the construction first attempts to match an idealized local template. For every coordination number greater than four, including those outside the named-template range, ligand-equivalence classes provide a general fallback. This is the practical realization of the “ XY_n local symmetry” idea: the algebra is local, the grouping is chemically typed, and the result does not mix angle rows with stretches or torsions.

7.2 Local Pseudosymmetry and Canonical SALCs

The molecular point group can be C_1 even when a methyl group, a local coordination sphere or a ring retains a useful approximate symmetry. SMITH therefore provides an optional local-pseudosymmetry construction before the exact molecular projector. For a bonded center, non-ring ligands are first partitioned by the continuous effective atomic number or synthon descriptor and by radial shell. The current validated defaults are 5×10^{-4} for the effective-atomic-number descriptor and 10^{-3} Å for the center–ligand distance. These are construction tolerances, not physical constants; they can be selected in the command-line

interface (CLI) or graphical user interface (GUI) and are written to the diagnostic record. The multiplicity pattern and ligand geometry then identify the local template, for example C_{3v} for a WXY_3 center or T_d for XY_4 . Coordination numbers 5–9 are treated by the same path. The current template pairs are trigonal-bipyramidal/square-pyramidal (coordination number, CN 5), octahedral/trigonal-prismatic (CN 6), pentagonal-bipyramidal/capped-octahedral (CN 7), square-antiprismatic/dodecahedral-like (CN 8), and tricapped-trigonal-prismatic/capped-square-antiprismatic (CN 9). If no template is available, the best score is outside threshold, or two scores are too close, the construction falls back to ligand-equivalence classes without changing the final rank criterion. If the sorted vectors of the $M = n(n - 1)/2$ ligand-pair cosines are $\mathbf{c}$ and $\mathbf{t}_T$, the template score is

$$s_T = \left[\frac{1}{M} \sum_{k=1}^M (c_k - t_{T,k})^2 \right]^{1/2}. \quad (3)$$

Sorting makes this score invariant to Cartesian orientation and ligand order. The best template is accepted only if $s_1 \leq s_{\max}$ and $s_2 - s_1 \geq \Delta_s$; the defaults are $s_{\max} = 0.12$ and $\Delta_s = 0.02$. The selected template, best score, margin, decision status and thresholds are frozen in the SONIC contract. The same coordinate identity is therefore retained in every use of the contract rather than reclassifying the center at each evaluation. The diagnostic additionally stores the signed headroom from both acceptance boundaries. Assignments within ten percent of a selected threshold are marked as near the root-mean-square (RMS) boundary, near the margin boundary, or near both. This warning does not rebuild the basis; it makes a potentially sensitive local decision visible before a geometry trajectory is started. Angles are partitioned separately according to the pair of ligand classes and, for an accepted high-coordination template, the ideal pair-cosine class, so that the SALC construction never mixes a stretch orbit with an angle orbit or two chemically different angular blocks.

For an ordered orbit of m equivalent primitive rows $p_1, \dots, p_m$, the first stored combination

is

$$q_{A_1} = \frac{1}{\sqrt{m}} \sum_{i=1}^m p_i. \quad (4)$$

The remaining deterministic Helmert rows are

$$q_{k+1} = \frac{\sum_{i=1}^k p_i - kp_{k+1}}{\sqrt{k(k+1)}}, \quad k = 1, \dots, m-1. \quad (5)$$

They form an orthonormal complement to q_{A_1} , have zero coefficient sum and fix a reproducible printed gauge. Unless a full local character table has also been resolved, the implementation deliberately labels these rows `NON_A1_1`, `NON_A1_2`, ... rather than assigning an unsupported local irrep. Thus “local A_1 ” is an ordering and partitioning property; the molecular irrep remains the one assigned later by the exact point-group projector.

For a ring, the candidate local operations are the rotations and reversals of the cyclic atom order. Only operations preserving the effective-atomic-number/ synthon coloring and the outgoing-bond signature are retained. The resulting cyclic subgroup is recorded with its operation count and confidence, while ring stretches, endocyclic valence angles and ring out-of-plane sources are transformed in separate blocks. In a fused system a shared edge is owned by the first ring stretch domain and reported as shared in the subsequent domain, avoiding a duplicate stretch source. **For an exocyclic central bond, equivalent primitive dihedrals are combined only inside an explicitly recognized XY_3 or acyclic XY_2 rotor domain. Their normalized collective coordinate is admitted only when the signed analytic Wilson row has nonzero rank, adequate norm stability and a continuous local reference branch. All other exocyclic bonds use one deterministic primitive dihedral; the failed domain or rank/stability test is recorded. Every generated row still passes the final analytic **B**-rank test.**

7.3 Exocyclic Torsions

For an acyclic bond $j-k$, all substituent pairs $i-j-k-l$ are possible primitive torsions. Keeping every pair is usually unnecessary and can create symmetry noise. Outside a special

rotor domain, the default **ONEDIH** rule therefore retains exactly one physical dihedral: the highest-priority admissible pair under effective-atomic-number and local-connectivity ordering, after linear-angle and small-ring exclusions. Repeated construction from the same frozen state selects the same ordered primitive; **ONEDIH** is not an orbit average.

For an equivalent $X(Y_1Y_2Y_3) - Z$ terminus, the special torsional coordinate is instead

$$q_{\text{rot}}^{XY_3} = \frac{1}{\sqrt{3N_W}} \sum_{w \in W(Z)} \sum_{a=1}^3 D(w, Z, X, Y_a), \quad (6)$$

where $W(Z)$ is the set of admissible non-linear skeleton anchors. The two differential branch directions of the Gaussian XY_3 span are represented in **SMITH** by the local bending block; Eq. 6 retains the common rotation, is normalized in coefficient space, and has periodicity three for an equivalent terminal XY_3 group. The acyclic XY_2 analogue is

$$q_{\text{rot}}^{XY_2} = \frac{1}{\sqrt{2N_W}} \sum_{w \in W(Z)} \sum_{a=1}^2 D(w, Z, X, Y_a), \quad (7)$$

with periodicity two. If equivalence, acyclicity, nonlinearity or an admissible anchor fails, construction falls back to the deterministic one-dihedral rule and records the reason. Finite torsional values use the stored reference branch or a circular $(\sin \phi, \cos \phi)$ representation, never an arithmetic mean across a 360° wrap.

7.4 Rings, Puckering and Butterfly Coordinates

Ring coordinates are the main place where primitive internal coordinates must be replaced by natural combinations. An n -membered ring has redundant endocyclic angle and out-of-plane source lists. **SONIC** constructs ring deformation coordinates and *RPck* puckering components as linear combinations of those source rows, keeping $n - 3$ independent puckering components for rings larger than three atoms. The endocyclic valence-angle combinations are the **CyGNA** source family and are stored as *RDef* rows; the corresponding endocyclic

out-of-plane combinations are stored as *RPck* rows. For each ordered ring stencil $D(a, b, c, d)$, the native source is the curvilinear coordinate $U(b, a, d, c)$; the dihedral only specifies the cyclic atom ordering. These signed U sources are the closest local curvilinear counterpart of the Cremer–Pople puckering displacements, while retaining analytic Wilson rows and a direct local definition for planar and nonplanar rings. This is a geometric analogy, not an identification: Cremer–Pople coordinates describe global ring-shape amplitudes and phases, whereas the (U) rows are local, signed curvilinear sources from which the active ($RPck$) combinations are constructed. This distinction is essential at a planar reference. A cyclic list of proper dihedrals is constrained by ring closure, and its branch phases and first-order rows can become linearly dependent when all ring atoms share a plane. The resulting failure belongs to the collective dihedral chart, even though an isolated proper dihedral need not itself be singular. By contrast, U is the signed displacement of one atom from a local three-atom plane: it is zero at planarity but has a finite, directly oriented Wilson row there. The native construction therefore projects the U -row space and verifies the rank of the complete Cartesian $\mathbf{B}$ after symmetry adaptation. The proper-dihedral stencil is retained only to define cyclic ordering, and an improper dihedral is introduced only by a terminal serializer that cannot represent U . The polar Cremer–Pople-like descriptors Q and ϕ are useful for interpretation, but the active coordinate basis stores linear *RPck* rows because their $\mathbf{B}$ rows are analytic linear combinations of out-of-plane derivatives.¹⁷ For every selected cosine–sine puckering pair, SMITH can also emit the derived observables

$$QPck_m = \sqrt{(RPck_m^c)^2 + (RPck_m^s)^2}, \quad \Phi P_m = \text{atan2}(RPck_m^s, RPck_m^c). \quad (8)$$

These $QPck/\Phi P$ variables are functions of the stored linear rows rather than additional rank-carrying coordinates.

The ring rows are a constructive extension of the natural-coordinate principle to cyclic source lists rather than a direct quotation of a published molecule-specific formula. If s_j

denotes such a list, for example the endocyclic valence angles or a curvilinear out-of-plane puckering list, the stored rows are real Fourier-like combinations

$$Q_m^c = N_m \sum_{j=1}^n s_j \cos \frac{2\pi m(j-1)}{n}, \quad Q_m^s = N_m \sum_{j=1}^n s_j \sin \frac{2\pi m(j-1)}{n}, \quad (9)$$

with the alternating row $Q_{n/2} = n^{-1/2} \sum_j (-1)^{j-1} s_j$ for even n . The normalization N_m is chosen so that coefficient vectors are unit length before the symmetry projection. Thus the same cyclic algebra is used for the CyGNA valence-angle family and for *RPck* out-of-plane puckering rows; only the source primitive list changes.

For a six-membered ring segment $1 \cdots 6$, a typical generated angle row has the explicit form

$$\begin{aligned} RDef = & 0.57735 A(6, 1, 2) - 0.28868 A(1, 2, 3) - 0.28868 A(2, 3, 4) \\ & + 0.57735 A(3, 4, 5) - 0.28868 A(4, 5, 6) - 0.28868 A(5, 6, 1), \end{aligned} \quad (10)$$

and the analogous puckering row is built from the corresponding cyclic U coordinates with the same normalized coefficient pattern. In the frozen contract these rows carry family labels, source primitive identifiers and symmetry labels; Q and ϕ are then functions of the *RPck* rows, not replacement active coordinates. Table 3 gives representative generated ring rows in a compact cyclic notation. Let $A_i = A(i-1, i, i+1)$ and $u_i = U(i+1, i, i+3, i+2)$, with atom indices taken modulo six. The printed signs are one valid gauge; for degenerate irreps a different orthonormal pair spans the same symmetry subspace.

Table 3: Representative six-membered-ring SONIC rows. Symmetry-preserving applications may select the totally symmetric subset; full-rank vibrational and sampling analyses retain the complete symmetry-labelled basis.

Stored row	Symmetry	Linear form
$A_{1g} RDef001$	A_{1g}	$(2A_1 - A_2 - A_3 + 2A_4 - A_5 - A_6) / \sqrt{12}$
$A_{2g} RPck001$	A_{2g}	$(2u_1 - u_2 - u_3 + 2u_4 - u_5 - u_6) / \sqrt{12}$
$E_u RPck001$	E_u	$(u_1 - u_2 - u_4 + u_5) / 2$
$E_u RPck002$	E_u	$(u_1 + u_2 - 2u_3 - u_4 - u_5 + 2u_6) / \sqrt{12}$

Unsaturated rings and locally rigid ring fragments require more than a generic $n - 3$ puckering count. The reconstruction method of Lema-Saavedra and Fernández-Ramos explicitly reduces the puckering dimensionality in the presence of rigid or multiple bonds and then recovers the full ring geometry by distributing the remaining distortions under constraints.¹⁶ The present SONIC construction already introduces bond-order-dependent flexibility factors in *RPck* source vectors, so out-of-plane sources associated with more rigid ring bonds contribute less to the puckering row. The rigid-bond treatment of Ref. 16 is therefore most naturally viewed in SONIC as a stronger, optional source-space contraction rather than as a rewrite of the covalent graph. A rigid bond $i - j$ can be replaced, inside that ring source polygon only, by the virtual source point $\mathbf{c}_{ij} = (\mathbf{r}_i + \mathbf{r}_j)/2$; the effective polygon then has one fewer source point and one fewer puckering degree of freedom. The analytic **B** row is expanded back to the real atoms by the chain rule, with the bond-center derivative distributed over i and j . This contraction is attractive for genuine double bonds, conjugated fragments and locally stiff fused-ring units, but it must be applied to complete symmetry orbits and only when the molecular-state continuous bond-order and synthon descriptors mark a hard local constraint. The present bond-order flexibility factors are therefore the conservative default; rigid-source contraction is the planned extension for cases where the chemically intended model removes, rather than merely down-weights, a non-puckering motion from the active source space. In both cases, the active GIC basis remains linear while Cremer–Pople amplitudes and phases are exposed as derived functions for sampling, PES fitting and conformer classification.

Fused, spiro and bridged systems are handled as connected ring-source spaces rather than as ordinary acyclic torsions. When two rings share a bond, SONIC adds butterfly coordinates around that shared bond. Condensed-ring torsions and ring puckering components remain separate homogeneous blocks in the symmetry projector. This is essential for high-symmetry systems such as cubane and for fused aromatic systems where a global torsion list would obscure the local ring role.

7.5 Out-of-Plane and Improper Coordinates

Three-coordinate centers generate native analytic out-of-plane coordinates unless all four atoms involved are cyclic. This U definition is used both for the primitive source and for the final SONIC coordinate; an improper dihedral is never selected during primitive generation or SMITH reduction. Only the Gaussian 16 .gjf writer translates $U(I, J, K, L) \mapsto D(J, I, L, K)$. Thus Gaussian compatibility is a terminal serialization step and cannot change the native coordinate model.

For a tricoordinate center, atom ordering is part of the native coordinate definition: in $U(I, J, K, L)$, I is the center, J and K define the local plane, and L is the displaced atom. Accordingly, an XY_2Z domain is written $U(X, Y_1, Y_2, Z)$, so that the equivalent pair defines the plane and the distinct substituent moves out of it. For three equivalent substituents, the complete oriented primitive orbit is retained and symmetry projected. When the three substituents are different, SMITH compares local C_{3v} -like and C_{2v} -like patterns: a clearly separated pair defines the plane in the latter case, whereas an ambiguous or C_{3v} -like environment uses the normalized oriented cyclic mean $\{U(I, J, K, L) + U(I, K, L, J) + U(I, L, J, K)\}/\sqrt{3}$. Its Wilson row is the same linear combination of the three analytic primitive rows. This choice is made once when the frozen contract is built and is never changed by a planarity threshold along an optimization, displacement, or inversion path; therefore the coordinate and its analytic derivative remain continuous through the planar geometry. As a final covariance audit, normalized analytic Wilson rows assigned to each irreducible representation must span an invariant subspace under every stored point-group operation within the stored tolerance.

8 Rank Reduction

The rank-reduction step takes the ordered primitive set $\mathcal{P} = (p_1, \dots, p_m)$, the analytic rows $\mathbf{b}(p_i)$ and a source-family map $F : \mathcal{P} \rightarrow \mathfrak{F}$. The family set $\mathfrak{F}$ carries a priority preorder $\preceq_{\mathfrak{F}}$.

Priority families—fragment coordinates, interaction-center distances, ring-puckering sources and related special objects—are minimal elements in this preorder; ordinary stretches, bends and torsions fill the rank only after those rows have been tested. Ties inside a family are resolved by topology, continuous-perception equivalence and CIP-like priority keys stored in $\mathcal{M}$. Thus the construction works on a fixed total order that extends a chemically motivated partial order.

Typed torsional-complement construction. The final torsional block is obtained as a non-null complement to the selected non-torsional space. Let $\mathbf{B}_N \in \mathbb{R}^{r_N \times 3N}$ contain the normalized Wilson rows already retained from all non-torsional families, and let $\mathbf{B}_T \in \mathbb{R}^{m_T \times 3N}$ contain the normalized candidate rows of one homogeneous torsional family. The Cartesian row-space projector associated with the non-torsional block and the residual torsional matrix are

$$\mathbf{P}_N = \mathbf{B}_N^\top (\mathbf{B}_N \mathbf{B}_N^\top)^+ \mathbf{B}_N, \quad \mathbf{R}_T = \mathbf{B}_T (\mathbf{I} - \mathbf{P}_N), \quad (11)$$

where the superscript $+$ denotes the Moore–Penrose inverse evaluated with the stored rank tolerance. Diagonalization of the residual metric gives

$$\mathbf{R}_T \mathbf{R}_T^\top \mathbf{U} = \mathbf{U} \boldsymbol{\Lambda}. \quad (12)$$

Columns of $\mathbf{U}_r$ are retained when $\lambda_i > \max(\tau_{\text{abs}}, \tau_{\text{rel}} \lambda_{\text{max}})$. The stored torsional coordinates and their Wilson rows are

$$\mathbf{q}_T^{\text{SONIC}} = \mathbf{U}_r^\top \mathbf{q}_T, \quad \mathbf{B}_T^{\text{SONIC}} = \mathbf{U}_r^\top \mathbf{B}_T. \quad (13)$$

The coefficient transformation acts exclusively inside the torsional family, and $\mathbf{U}_r^\top \mathbf{R}_T$ has full row rank relative to the selected non-torsional span. Pseudo-edges, fused-ring closures, and other topologies that generate excess torsional candidates therefore yield exactly $\text{rank}(\mathbf{R}_T)$ typed torsional coordinates. For a nontrivial molecular point group, the construction is applied

to homogeneous torsional source blocks using the stored atom permutations and orientation signs; the established block-local symmetry projector then resolves the retained subspace into irreducible representations. Coordinate families are never mixed by this operation.

For the formic-acid–water pseudobond fixture, ten candidate torsional rows produce six non-null complementary coordinates and four null combinations. The complete Wilson-matrix condition number changes from approximately 91 to 53, the target vibrational rank is retained, and the same construction passes the nontrivial-point-group covariance fixtures.

The following design invariants are consequences of that ordered protocol. They are stated explicitly to make the construction verifiable, not to claim uniqueness over all possible internal-coordinate bases.

Relabelling covariance. Let two frozen molecular states be related by an atom relabelling π that preserves element identities, geometry, topology, perception descriptors, fragments, interaction centers, point-group operations, priority keys and tolerances. Let Q_π be the induced orthogonal permutation of Cartesian displacement coordinates and U_π the induced permutation/sign map on primitive coordinate rows. If SONIC returns a contract $\mathcal{C}$ for the first state, then it returns an isomorphic contract $\pi_*\mathcal{C}$ for the second state. In particular,

$$\mathbf{B}(\pi_*\mathcal{P}) = U_\pi \mathbf{B}(\mathcal{P}) Q_\pi^T,$$

and the accepted coordinate spaces, family labels and symmetry-adapted blocks are carried into each other by the induced row and coefficient-space isomorphisms. This follows because every primitive generator, row norm, rank residual and homogeneous-block projector is evaluated from data preserved by π ; orthogonal permutation/sign transformations do not change the Boolean acceptance tests.

Equivalently, SONIC is natural with respect to symmetry-preserving atom relabellings: relabelling the molecular state and then constructing the contract gives the same object, up to the canonical contract isomorphism, as constructing the contract first and then transporting

it by the relabelling.

Reproducible contract. For a fixed molecular state–geometry, topology, perception descriptors, point-group operations and fragment data—and for fixed primitive generators, family priority order, tie-breaking keys, row-normalization rule and numerical tolerances, SONIC returns a unique contract, or a unique incompatibility certificate. In particular, no stochastic search, optimizer history or manual choice is part of the contract definition. The verification is direct: each stage is a fixed map of the frozen input data, and the fixed order of candidate rows converts every rank and symmetry decision into a stored threshold test.

Semantic preservation. Let $\mathcal{P}_{\text{prot}}$ be the priority prefix of the ordered candidate set. Every candidate in this prefix whose Wilson row is nonsingular and independent, within the stored tolerance, of the previously accepted prefix rows is retained in any returned final object. No ordinary stretch, bend or torsion can displace such an accepted coordinate. If the independent prefix already exceeds the target vibrational rank, SONIC reports an invalid model rather than removing a semantic coordinate by exchange. This follows directly from the ordered-prefix construction.

Rank-preserving selection. Assume exact arithmetic and define

$$\mathcal{R}_k = \text{span}\{\mathbf{b}(p_1), \dots, \mathbf{b}(p_k)\}.$$

Incremental Gram–Schmidt selection accepts p_k exactly when $\mathbf{b}(p_k) \notin \mathcal{R}_{k-1}$. After all candidates have been tested, the accepted subset $\mathcal{A}$ is an ordered basis of $\mathcal{R}(\mathcal{P})$. If $\text{rank } \mathbf{B}(\mathcal{P}) = r$, then $|\mathcal{A}| = r$ and $\text{span } \mathbf{B}(\mathcal{A}) = \text{span } \mathbf{B}(\mathcal{P})$. Indeed, at step k Gram–Schmidt decomposes $\mathbf{b}(p_k)$ into its projection on the current accepted span plus an orthogonal residual; a zero residual leaves the span unchanged, and a nonzero residual adds exactly one new independent direction.

The numerical construction is the finite-precision version of this statement. SONIC selects rows by incremental modified Gram–Schmidt rank testing with a fixed numerical tolerance, but the comparison is performed on normalized Wilson rows,

$$\hat{\mathbf{b}}(p) = \frac{\mathbf{b}(p)}{\|\mathbf{b}(p)\|_2}. \quad (14)$$

The residual tested for candidate p is

$$\rho(p) = \left\| \hat{\mathbf{b}}(p) - \sum_{i \in \mathcal{A}} [\hat{\mathbf{b}}(p) \cdot \hat{\mathbf{b}}_i] \hat{\mathbf{b}}_i \right\|_2, \quad (15)$$

where $\mathcal{A}$ is the current accepted orthonormal row basis. The absolute row norm is used only to reject singular coordinates; independence is decided by the direction of the analytic Wilson row. This normalization avoids a direct comparison of raw angstrom and radian scales in the final rank test. It does not remove all modelling choices: the ordered candidate list remains a chemical convention, not a uniqueness guarantee. Singular rows and rows dependent on previously accepted rows are recorded in the frozen diagnostics.

The decisive policy is therefore not only the linear algebra routine; it is the priority constraint under which the basis is built. Special coordinates are tested before ordinary primitives. Ordinary stretches, bends and torsions then complete the rank. A priority coordinate can be skipped only if it is singular or dependent on previously accepted priority rows. It cannot be displaced merely because an ordinary valence coordinate spans a similar first-order motion. If these rows alone exceed the vibrational rank, the construction is rejected rather than silently discarding a special coordinate.

Block-local scaling invariance. Let $H \subset \mathcal{P}$ be a homogeneous coordinate block. Uniformly rescaling all Wilson rows in that block by a nonzero scalar does not change the normalized-row rank decisions inside H , nor the singular-vector or projector subspaces obtained from $\mathbf{B}_H \mathbf{B}_H^T$. By contrast, a global decomposition of several heterogeneous blocks is

generally not invariant to independent rescalings of those blocks. The reason is that row normalization removes a uniform factor inside H , and $(\lambda \mathbf{B}_H)(\lambda \mathbf{B}_H)^T = \lambda^2 \mathbf{B}_H \mathbf{B}_H^T$ rescales eigenvalues but not eigenspaces; concatenating heterogeneous blocks before diagonalization changes their relative weights.

Local stability. Let S_0 be a nonsingular frozen molecular state. Assume that, in a neighbourhood of S_0 , topology, fragment decomposition, perception equivalence classes, point-group representation, priority prefix, typed block membership and priority order remain fixed in a ball of radius δ_{str} around S_0 . Let $\{f_a(\mathbf{x})\}_{a=1}^{n_t}$ be the finite set of scalar decision functions used by the construction in this stratum: row norms, rank residuals, block singular or projected-vector norms and any sign-gauge pivot quantities used to store SALC coefficients. Let τ_a be the corresponding acceptance threshold and define the margin at S_0

$$\mu_a = |f_a(\mathbf{x}_0) - \tau_a| > 0.$$

Assume that each f_a is Lipschitz on the whole stratum ball with a uniform constant L_a , i.e.,

$$|f_a(\mathbf{x}) - f_a(\mathbf{y})| \leq L_a \|\mathbf{x} - \mathbf{y}\|$$

for all $\mathbf{x}, \mathbf{y}$ in that ball. Then every

$$0 < \delta < \min \left[\delta_{\text{str}}, \min_{a: L_a > 0} \frac{\mu_a}{2L_a} \right]$$

preserves the discrete contract, with the usual convention that the inner minimum is $+\infty$ if all $L_a = 0$. Thus every state S in the same discrete stratum with $\|\mathbf{x}(S) - \mathbf{x}(S_0)\| < \delta$ generates an isomorphic contract: the primitive identities, accepted/rejected rank decisions, priority coordinates, family labels and symmetry labels are unchanged, while $\mathbf{B}$ rows and coefficient matrices vary continuously with the geometry under the same normalization convention. This

is the usual finite-decision margin argument: each decision function stays on the same side of its threshold for perturbations smaller than $\mu_a/(2L_a)$, and the minimum over the finite decision set preserves the entire ordered construction.

This is the technical reason for block-local reduction. SONIC does not claim that weights disappear from all possible coordinate constructions. It avoids using arbitrary cross-family weights to define chemically labelled variables. Only after typed blocks have been reduced and special coordinates have been accepted does the final normalized rank audit compare directions across families.

The local-stability result is deliberately stratum-local rather than global. A change in topology, symmetry, local equivalence or a near-threshold residual is a genuine model change; it is therefore reported as contract provenance rather than hidden behind a re-diagonalized global basis.

This policy is what makes the coordinate set suitable for refinement. A fragment-center distance or ring-center to atom distance might be expressible as a linear combination of many ordinary internal derivatives near a reference geometry. That algebraic fact does not make the ordinary combination a useful fit variable. The priority-first rule preserves the variables that define the model, then uses ordinary coordinates for completeness.

9 Symmetry Adaptation

Two symmetry levels are kept distinct. The local construction described above uses center, ring or bond domains to obtain chemically ordered source blocks even for a globally asymmetric molecule. Exact molecular symmetry is applied only after a non-redundant coordinate basis has been selected and remains the final authority for molecular irrep labels and totally symmetric export. Let G_{pt} be the molecular point group stored in the source state. The point group is not guessed from idealized element labels alone: the molecular-state builder first clusters atoms by continuous element, coordination, bond-order and synthon descriptors,

then tests candidate rotations, reflections and inversion operations against the Cartesian geometry within the stored tolerances. An operation is accepted only when it maps each atom onto an atom of the same continuous equivalence class and preserves the stored topology and local descriptor signatures. The accepted operations, atom permutation table, tolerances and perception version are frozen before SONIC constructs coordinates. For a path whose symmetry changes, the user may request a reduced working subgroup, for example C_1 , C_s , C_i or C_2 . SONIC then filters the stored operations to that subgroup and uses the reduced group consistently for labels, projectors and totally symmetric coordinate export. Each operation $h \in G_{\text{pt}}$ is represented by a Cartesian rotation matrix, an atom permutation and the signs required for oriented coordinates such as torsions, out-of-plane bends and axial fragment rotations. For a homogeneous accepted coordinate block H , with coefficient space V_H , the stored operations map the block into itself. **Before any projector is formed, the frozen finite action is audited for identity, the order expected from the declared group, inverses, and closure of both its Cartesian matrices and atom permutations. This audit is applied to the complete supported finite-group families $C_n, C_{nv}, C_{nh}, D_n, D_{nd}, D_{nh}$ for $n = 2, \dots, 6$, $S_4, \dots, S_{12}$, C_1, C_s, C_i , and the tetrahedral, octahedral and icosahedral families; linear groups are handled separately and are not forced into a fictitious finite closure. Whether the operation table is supplied by ORACLE or built by the standalone frontend, the same matrices and permutations determine the character sequence used by SMITH. No independent name-based reconstruction or lower-order surrogate subgroup is introduced, so high-order groups and degenerate SALC blocks are treated by the same general construction. Homogeneous blocks admitted to the projector are invariant under this permutation and sign representation; their coefficient spaces therefore carry finite-dimensional representations of the point group,**

$$\Gamma_H : G_{\text{pt}} \rightarrow GL(V_H).$$

In the simplest case $\Gamma_H(h)$ permutes equivalent coordinate basis vectors and multiplies them by the appropriate parity sign.

If H is closed under the stored operations, the symmetry-adapted subspace of irreducible representation α is obtained by the standard character projector

$$\Pi_\alpha^{(H)} = \frac{\ell_\alpha}{|G_{\text{pt}}|} \sum_{h \in G_{\text{pt}}} \chi_\alpha(h)^* \Gamma_H(h), \quad (16)$$

where ℓ_α is the dimension of α and χ_α its character. SONIC applies these projectors to the coefficient space of the block, orthonormalizes the nonzero projected vectors and stores the resulting SALC matrix $C_{\alpha H}$. The symmetry-adapted Wilson rows are then

$$\mathbf{B}_{\alpha H} = C_{\alpha H} \mathbf{B}_H.$$

For tasks that must preserve the stored point-group symmetry, SONIC can export only the totally symmetric rows, such as A_1 , A' or A_{1g} depending on the group. This is the natural subset for symmetry-preserving geometry optimizations and semiexperimental structure fits. The remaining irreps are still retained in the full contract for rank closure, GF/PED block analysis, DVR coordinates and diagnostics. If an irreducible representation occurs with multiplicity greater than one in the same homogeneous block, the projector defines a subspace, not a unique printed basis. SONIC fixes the stored basis by applying modified Gram–Schmidt to projected source vectors in the canonical block order, by discarding vectors below the saved norm tolerance, and by choosing the sign from the largest-magnitude coefficient pivot. The subspace is the representation theoretic object; the ordered, sign-gauged coefficient matrix is the deterministic serialization stored in the contract. Along scans or optimizations, a change in the identity of the largest pivot can flip the printed gauge even when the invariant subspace, rank and symmetry labels are unchanged. SONIC therefore treats pivot changes as executable path diagnostics: consecutive SALC coefficient matrices are compared by subspace overlap, pivot changes are flagged as gauge events, and the same diagnostic module provides

a Procrustes alignment for downstream GF/PED or DVR layers when a path-continuous row basis is required. None of the reported chair-to-boat, ferrocene or GF/PED probes showed a pivot switch in the active blocks used for the tabulated quantities.

The block-local rule is the key safeguard. A stretch and a torsion may belong to the same irreducible representation, but they are not the same kind of coordinate. Mixing them during symmetry adaptation would produce a valid vector in an abstract representation and an opaque variable in a chemical model. SONIC therefore projects only within coordinate families and source blocks. If a block is not closed under the stored operations, the construction uses an explicit local SALC fallback only where the contract allows it, and records that fallback in the symmetry diagnostics.

For a multi-fragment system, independent blockwise choices can nevertheless be collectively rank deficient even when every projected candidate is individually valid. Resolving such a dependence by taking the first available row would make the result depend on family order, whereas a global SVD would restore rank only by mixing chemically different coordinate types. SONIC instead treats the projected candidates as the ground set of two matroids. Linear independence of their analytic Wilson rows defines a linear matroid, while the requirement to retain exactly the native dimension of every coordinate family defines a partition matroid. An augmenting-path matroid-intersection selection finds a common basis with full molecular rank and the exact family quotas; subsequent conditioning exchanges are allowed only within the same family.¹⁸ If no such common basis exists, construction fails rather than silently changing the coordinate model. The accepted basis is finally audited under every stored point-group operation to verify both irrep covariance and cross-family rank.

Fragment translations transform as polar vectors; fragment orientations transform as axial vectors. Torsions and out-of-plane coordinates carry the appropriate sign changes under atom permutation and inversion. Ring puckering coordinates are projected through the Cartesian row space of the native oriented U sources. Their source and target phases are both retained under each atom permutation, and a final global rank test prevents blockwise symmetry

adaptation from introducing a cross-family dependence. Thus the symmetry labels refer to the same analytic **B** rows serialized in the frozen contract.

10 Intermolecular and Special Coordinates

Weak complexes are a stress test for any internal-coordinate generator. A covalent graph alone is disconnected, but adding arbitrary weak contacts as ordinary bonds can create artificial rings and mix covalent and non-covalent motions. Within its current validated scope, SONIC therefore keeps two explicit intermolecular constructions: a fragment-coordinate model and a pseudo-bond/pseudo-cycle model for selected hydrogen-bonded or contact-closed graphs.

In the release-grade `SPECIAL_COORDINATES` mode, each fragment keeps its intramolecular coordinate basis. Interfragment motion is represented by special coordinates: fragment-center distances, fragment-center to atom distances, Cartesian components of relative fragment translation and relative fragment orientations expressed as exponential-map rotation coordinates. This construction explicitly adopts the fragment-rotation parametrization introduced by Wang and Song in TRIC/geomeTRIC: collective translations are described through fragment centroids, while orientations are represented by an exponential map of quaternions.¹⁰ The use of this exponential mapping in SONIC should therefore be credited to the geomeTRIC work.

This shared parametrization does not make the two coordinate constructions identical. In TRIC/geomeTRIC, three translations and three rotations are added for each fragment and are subsequently combined with the intrafragment primitives in a delocalized coordinate space.¹⁰ Within the SONIC contract, one fragment is instead selected deterministically as the reference; three relative translations and, for non-linear partners, three relative exponential-map orientations are generated for every other fragment, so that the overall rigid translation and rotation of the complete complex do not enter the molecular vibrational contract. These rows remain a protected semantic block during rank reduction rather than being delocalized with stretches, bends and torsions. They transform explicitly as polar and axial vectors under

the stored point-group operations and are serialized together with their centers, body frames, gauge choices, symmetry labels and analytic **B** rows. Thus SONIC adopts and credits the TRIC exponential map while embedding it in a distinct, frozen coordinate construction.

Fragment coordinates are not the only intermolecular representation available in SONIC. For a weak complex, the coordinate contract can select either the `SPECIAL_COORDINATES` fragment construction just described or the `PSEUDO_BONDS` construction. The former preserves the disconnected monomers and represents their relative rigid-body motion explicitly; the latter inserts selected weak contacts as typed pseudo-edges and generates the associated pseudo-cycle coordinates without reclassifying those contacts as covalent bonds. The selected construction is recorded in the contract, so the two descriptions remain explicit alternatives rather than indistinguishable members of one primitive pool.

Fragment centers, body frames and orientation-chart references are serialized with the coordinate definition. Relative orientations use the exponential-map chart on $SO(3)$, with the same finite-rotation parametrization as TRIC/geomeTRIC, but SMITH does not prescribe how a requested finite coordinate change is converted back to Cartesian geometry. Choice of right inverse, damping, finite rotations, continuation, chart rebasing, “soft” and “hard” partitions, symmetry maintenance and backend-frame alignment all belong to the application using the contract. The SMITH motion files discussed below are explicitly first-order visualizations at fixed **B**, not general internal-to-Cartesian solvers.

Interaction centers extend the same idea to non-atomic reference objects. A center may be a ring centroid, a bond midpoint or a contact center. Distances from atoms or fragments to such centers are special coordinates and are evaluated analytically by distributing the center derivative over the atoms that define it. This gives a practical route to coordination complexes, metallocenes and eta-like interactions without pretending that a metal-ring centroid is a covalent atom in the molecular graph.

The molecular structures behind these constructions are collected in Fig. 3. This dedicated view distinguishes stored centers and center/contact rows from the ordinary covalent skeleton

and makes explicit that special coordinates are exercised for metallocenes, hydrogen-bonded dimers, aromatic-polar complexes and a supplied η^3 ligand center.

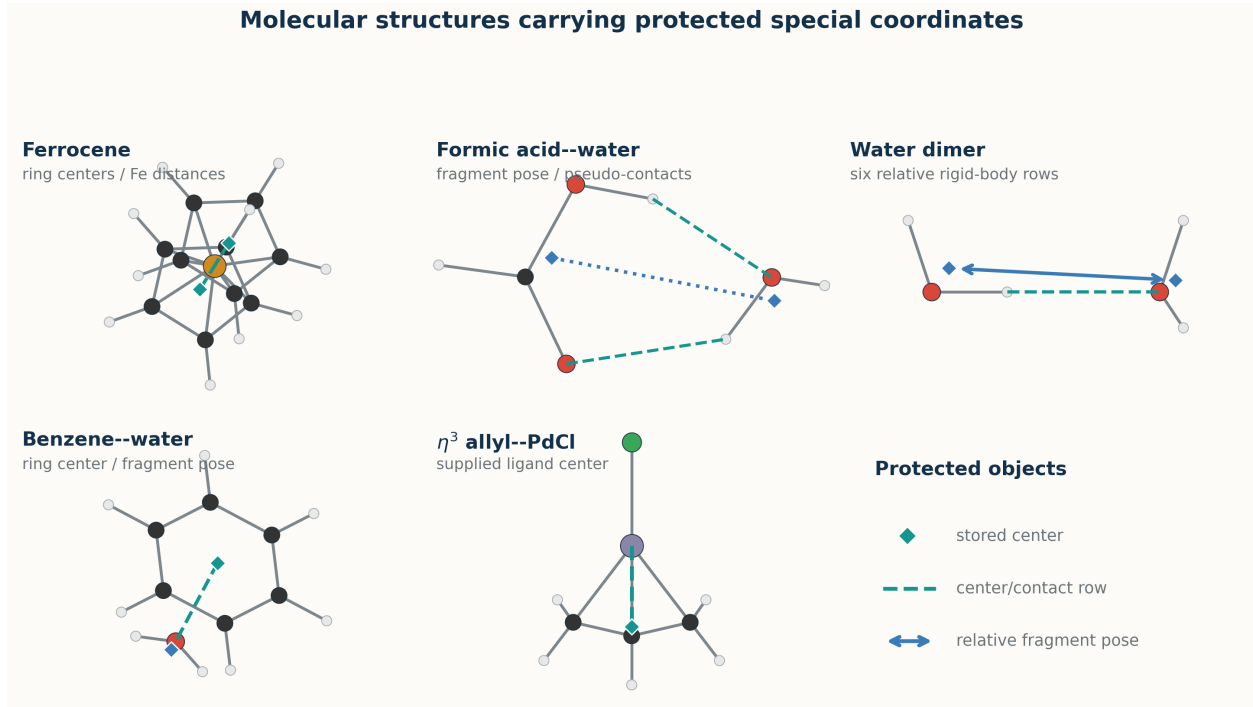


Figure 3: Protected special-coordinate fixtures. Diamonds mark ring, ligand or fragment centers; dashed teal segments mark center-atom or contact coordinates; blue elements mark relative fragment pose. The η^3 complex retains its supplied center without a fictitious covalent atom.

Ferrocene provides a compact example. If the two cyclopentadienyl ring centers are

$$C_{031} = \frac{1}{5} \sum_{i \in R_1} \mathbf{x}_i, \quad C_{032} = \frac{1}{5} \sum_{i \in R_2} \mathbf{x}_i,$$

and the iron atom is atom 2, SMITH first defines the two center-atom distances

$$d_1 = |C_{031} - \mathbf{x}_2|, \quad d_2 = |C_{032} - \mathbf{x}_2|.$$

The protected symmetry-adapted rows retained in the contract are then

$$A'_1 CnAt001 = \frac{d_1 + d_2}{\sqrt{2}}, \quad A''_2 CnAt001 = \frac{d_1 - d_2}{\sqrt{2}}.$$

The A'_1 row is the total-symmetric metal–ring breathing coordinate; the A''_2 partner is kept in the full contract and marked frozen when only total-symmetric coordinates are requested for an optimization or semi-experimental fit. In Gaussian syntax the centers are serialized as inactive Cartesian functions such as

$$CxC031 = (X_1 + X_3 + X_6 + X_5 + X_4)/5,$$

and the active distance row is written as the corresponding square-root function of the center coordinates and the iron Cartesian coordinates. The special coordinate is therefore generated from the molecular model, not hand-entered as an extra atom.

The η^3 allyl–palladium standalone probe exercises the same interface with a non-ring ligand center. For allyl carbon atoms 3–5, the input state supplies

$$C_{001} = \frac{\mathbf{x}_3 + \mathbf{x}_4 + \mathbf{x}_5}{3}$$

with center kind `ETA3_CENTER`; `SMITH` then protects

$$q_{\eta^3} = |C_{001} - \mathbf{x}_{\text{Pd}}|.$$

The idealized ten-atom fixture closes at rank 24/24, and the reference value is $q_{\eta^3} = 1.8000 \text{ \AA}$. Its Gaussian 16 export writes the three center components as inactive Cartesian functions and the metal–center distance as an active square-root expression. This is deliberately an interface and rank-closure test rather than an optimized electronic-structure benchmark: definition of the η^3 center belongs to the input provider, whereas its retention, first differentiation and serialization belong to `SMITH`.

The same contract applies without case-specific rules to a general η^n interaction. When a canonical metal–ligand center is present, the active SONIC source contains one metal–center coordinate rather than a redundant fan of metal–member distances. The individual contacts

may remain available as chemical diagnostics, but they do not generate duplicate active stretches, bends or torsions. A downstream local force-field or Lindh-type guess Hessian first constructs its interaction graph with the corresponding pseudo-bonds and then suppresses the metal–member valence terms represented by the protected center coordinate; the η^n curvature is therefore counted once. The rule covers η^3 , η^5 , and higher hapticities uniformly, whereas a quantum-mechanical Hessian is transformed directly in the coordinate basis selected for the optimization.

In the current validated PSEUDO_BONDS baseline, selected non-covalent contacts are inserted into the construction graph as typed pseudo-edges, and the resulting pseudo-cycles generate angular and torsional source components analogous to covalent ring coordinates. These rows are still labelled separately from covalent stretches, bends and torsions, so downstream users can decide whether they are model coordinates, constraints or diagnostics. The present validation uses formic-acid–water, formamide–water and the glycine I/II hydrogen-bond fixtures as regression probes: a single intramolecular bridge and a bifurcated bridge both close the expected rank without converting the underlying covalent topology into a different molecule.

A concrete non-covalent example is the formic-acid–water dimer used in the regression suite. In pseudo-bond mode, the validated weak contacts (3,8) and (5,6) are inserted as typed HBOND pseudo-edges. They are stored in the contract as

```
PSEUDO_BOND_COUNT 2,      1 3 8 KIND=HBOND,      2 5 6 KIND=HBOND.
```

The augmented graph then produces ordinary stretch, bend and torsion candidates plus protected pseudo-cycle bend and torsion families. The final object remains rank-complete at 18/18, and the pseudo-cycle rows remain labelled separately from covalent ring coordinates so that downstream analyses can keep or suppress them explicitly.

Additional pseudo-bond and intermolecular fixtures were added to keep this baseline

close to the weak-complex cases used by geomeTRIC-style translation-rotation coordinates (Table 4). Here and below, point group (PG) refers to the exact molecular point group. The point of the table is deliberately modest: these systems verify that typed contacts, fragment-center variables and pseudo-cycle rows can be constructed and labelled without collapsing the fragment model into a covalent molecule. They do not yet establish full generality for arbitrary non-covalent networks.

Table 4: Weak-complex and pseudo-cycle fixtures used to test coexistence of fragment, contact and point-group models.

System	PG	Coordinate mode	Diagnostic role
Formic acid-water	C_s	typed H-bond pseudo-edges and fragment centers	two weak contacts close a hydrogen-bonded pseudo-cycle while preserving the monomer fragments.
Water dimer	C_s	protected fragment translations, orientations and center distances	geomeTRIC-like weak-complex rotor proxy; tests that orientation rows have nonzero sparse derivatives in a non-covalent dimer.
Li ₂ -Ar	linear/axial	protected interfragment axes plus polar and elliptic functions	Gaussian-GIC scan motif generated from fragment translation axes rather than hand-written coordinate expressions.
Borane-ammonia	C_{3v}	fragment-center distance and axial relative orientation	nontrivial symmetric donor-acceptor complex; tests vector and axial-vector transformation of fragment variables.
Glycine I	C_1	intramolecular H-bond pseudo-edge	single internal weak bridge; tests pseudo-cycle construction without intermolecular fragments.
Glycine II	C_1	bifurcated intramolecular H-bond pseudo-edges	multi-contact weak bridge; tests deterministic contact ordering and labelled pseudo-cycle rows.

11 Validated Construction Scope

Table 5 records the status of the present SONIC construction. The purpose is not to document a program interface, but to state which mathematical objects belong to the present contract and which objects remain methodological extensions. This separation is important because downstream validation is meaningful only for coordinates that are part of the frozen contract.

The boundary at second order deserves an explicit statement. The first Wilson-matrix derivative is

$$\mathcal{D}_{iab} = \frac{\partial B_{ia}}{\partial x_b}. \quad (17)$$

Table 5: Validated scope of the main SONIC coordinate families.

Feature	Status	Notes
Covalent stretches	Active	All topology bonds; local equivalence grouping available.
Exocyclic bends	Active	Bonded-center bends; local symmetric combinations for common motifs.
High coordination	Active	Two named templates per CN 5–9; generic ligand-equivalence fallback for every CN > 4. Python is dynamically sized; the distributed Fortran backend is compiled through CN 32.
Domain-first ownership	Active	Stretches are followed by complete ring, XY_3 , acyclic XY_2 , fragment and semantic special blocks; ordinary coordinates are generated only from residual unowned primitives.
Collective XY_3/XY_2 rotors	Active optional path	Normalized common-rotation coordinates with threefold/twofold periodicity; branch modes remain in owned bending/out-of-plane blocks; ambiguous domains fall back conservatively.
One-dihedral exocyclic torsions	Active default	ONEDIH retains exactly one deterministic physical dihedral for each eligible non-special bond; it is not an averaged orbit.
Ring deformation/puckering	Active	Endocyclic angle and torsion sources generate <i>RDef</i> and <i>RPck</i> ; Q/ϕ are derived descriptors.
Butterfly/condensed rings	Active	Fused-ring butterfly and condensed torsion families remain separate homogeneous source blocks.
Out-of-plane rows	Active	Native U primitives and SONICs; improper dihedrals are generated only in a requested Gaussian 16 .gjf export.
Special fragment coordinates	Active	Fragment-center distances, center-atom distances, translations and orientations are protected first.
Interaction centers	Active baseline	Ring, bond and related virtual centers can enter center-atom coordinates.
Semantic PROTECT/OBSERVABLE layer	Active baseline	Deduplication, row-equivalent substitution, rank-excess rejection and sparse observable propagation.
Pseudo-bond/pseudo-cycle mode	Validated baseline	Typed weak contacts become pseudo-edges; pseudo-cycle bends/torsions stay labelled separately.
Point-group projector	Active	Applied after non-redundant reduction inside homogeneous blocks; fallback is recorded.
Local-pseudosymmetry SALCs	Active optional path	Center, ring and bond domains use effective-atomic-number/synthon and radial equivalence; the local A_1 row is first and the exact molecular projector remains final.
SONIC symmetry-adapted Cartesian reference	Active	Cartesian active-space reference for validation; does not change GIC construction.
Sparse Wilson- B representation	Validated baseline	Sparse derivative rows are stored; dense matrices are only audit representations.
First derivative of B	Outside the SMITH contract	B' is not required to construct SONIC and is neither built nor serialized by standalone SMITH. A second-order consumer may evaluate it on demand from the frozen coordinate definitions.
Primitive/function registry	Validated baseline	Canonical primitive and $\mathbf{y} = f(\mathbf{q})$ records prevent duplicate coordinate objects.
Coordinate-function layer	Validated baseline	Inverse/exponential distances, trigonometric functions, polar radii and quadratic functions use sparse chain rules.

This object is often denoted $\mathbf{B}'$, but it is not needed to define a coordinate, to select the non-redundant SONIC basis or to evaluate its Wilson matrix. Accordingly, SMITH does not construct or store it. A program that must transform a Hessian at a nonstationary geometry is responsible for evaluating $\mathcal{D}$ on demand from the frozen primitive definitions and SONIC coefficients. Such an evaluator may assemble analytic primitive Cartesian Hessians into sparse slices and use finite differences only as a reported audit or fallback, but that is a second-order application service, not part of the published SMITH interface.

The reason an application may need this service is the exact Hessian chain rule. At a nonstationary geometry the coordinate-curvature contribution is

$$C_{ab} = \sum_i g_i^{(q)} \mathcal{D}_{iab}, \quad (18)$$

and symmetrized only after the contraction. If $\mathbf{g}^{(q)}$ and $\mathbf{H}^{(q)}$ are the internal-coordinate gradient and Hessian, the exact Cartesian derivatives are

$$\mathbf{g}^{(x)} = \mathbf{B}^T \mathbf{g}^{(q)}, \quad (19)$$

$$\mathbf{H}^{(x)} = \mathbf{B}^T \mathbf{H}^{(q)} \mathbf{B} + \mathbf{C}. \quad (20)$$

Conversely, on the full-row-rank active coordinate space,

$$\mathbf{H}^{(q)} = (\mathbf{B}^+)^T (\mathbf{H}^{(x)} - \mathbf{C}) \mathbf{B}^+. \quad (21)$$

Thus $\mathbf{B}'$ is not an auxiliary coordinate-selection device: it is the geometric-curvature term required to transform Hessians at a nonstationary or constrained geometry. At an unconstrained stationary point $\mathbf{g}^{(q)} = 0$, so $\mathbf{C} = 0$ and the familiar first-order transformation is recovered. The distinction matters for scans, constrained optimizations, force-field construction away from equilibrium and any trajectory on which Cartesian and internal Hessians are exchanged. It does not enlarge the SONIC contract: the consumer builds only the derivative slices it

needs and remains responsible for singular-chart handling and numerical fallback policy.

12 Validation Strategy

The validation target is not merely that a coordinate list can be printed. A SONIC coordinate definition must first pass construction-level checks. The final coordinate count and analytic **B** rank must match the vibrational rank. The current Python construction and a separate Fortran control implementation must agree on coordinate families, final rank and Wilson-**B** row space for the golden corpus. Symmetry-adapted blocks must close under the stored point-group operations, and the saved SALC coefficient vectors must remain normalized. These are internal implementation-level checks, not a comparison with an external published coordinate generator.

The broader scientific motivation includes vibrational analysis, sampling and semi-experimental refinement, but the validation layers must be separated. The present paper validates the coordinate contract itself and tests whether an external Gaussian 16 calculation can consume SONIC-generated coordinates. Table 6 summarizes the claim supported by each evidence layer. The scope of the numerical evidence retained here is stated explicitly in Sec. 14.2.

Figure 4 shows representative molecular systems used in the validation probes. The figure is included to keep the coordinate discussion tied to the underlying chemistry: bridged hydrocarbons test ring and improper-coordinate choices, camphor adds a rigid asymmetric framework, ferrocene exercises special center coordinates, the formic-acid–water complex tests protected interfragment translations and orientations, the η^3 allyl–palladium complex tests a supplied ligand center, and the cyclohexane sequence tests portable puckering paths.

The regression corpus contains acyclic and cyclic molecules, fused and non-benzenoid rings, bridged and spiro systems, high-symmetry cages, linear molecules, ferrocene in D_{5h} and D_{5d} forms, a formic-acid–water weak-complex probe, and an η^3 allyl–palladium interaction-

Table 6: Validation map separating scientific claims from implementation diagnostics.

Evidence	What it tests	What it does not claim
Internal corpus, Table 7	Rank closure, analytic B -row consistency, point-group block closure and agreement with a separate Fortran control implementation.	External superiority over another published coordinate generator.
Local-pseudosymmetry corpus, Table 8	Local A_1 -first ordering, rigid-rotation invariance, substituted/fused-ring domain records and independent Python/Fortran classification.	Replacement of the exact molecular point group or a complete local-irrep decomposition.
Gaussian-16-GIC-inspired motifs, Table 9	Automatic construction of selected coordinate objects related to examples in the recent GIC work.	Use or validation of its development-version implementation, or a benchmark of the Gaussian 16 internal optimizer.
Gaussian 16 Rev. C.01 ReadAllGIC optimizations, Table 10	Whether the released parser and optimizer can consume the generated coordinate block.	A universal iteration-count advantage for every molecule.
Puckering and portable-export probes, Figs. 6 and 7, with cross-code values in SI Sec. S5	Reuse of the same generated coordinate path as Gaussian GIC input or ordinary Cartesian geometries for other electronic-structure codes.	A full DVR application.
Construction-cost study, Table 12 and Fig. 8	Empirical scaling of coordinate construction and sparse- B evaluation on the present test corpus.	Asymptotic proof for all molecular graph families.

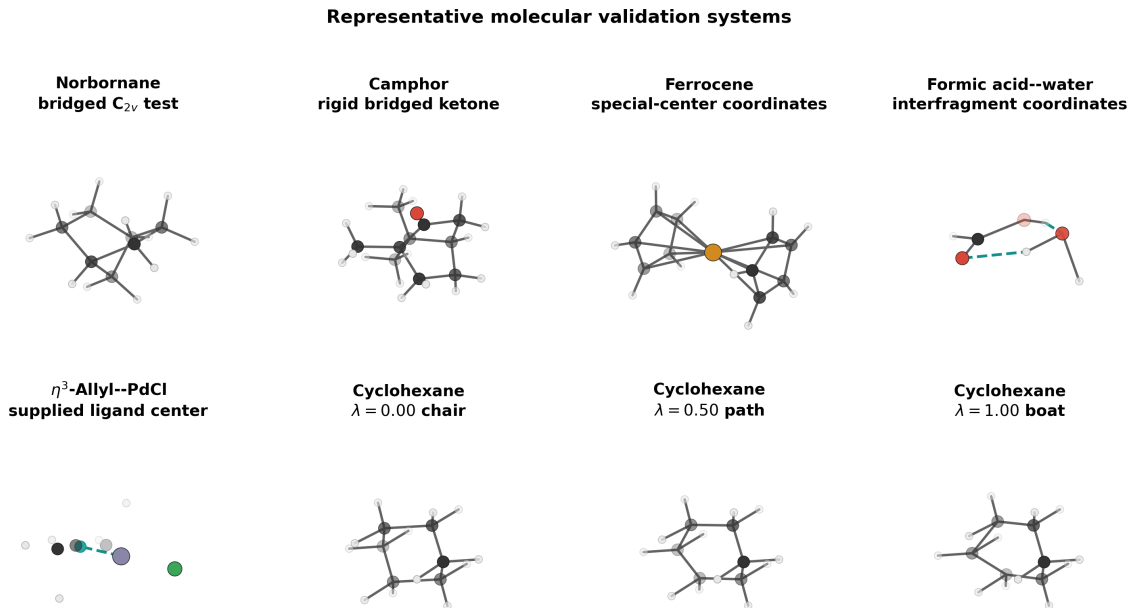


Figure 4: Representative validation systems for bridged rings, center-based coordinates, relative fragment pose and a protected η^3 ligand center. Dashed turquoise segments mark contacts or center-metal coordinates; the cyclohexane panels sample the chair-to-boat path.

center probe. The ring subset protects the *RDef/ RPck* source spaces, butterfly coordinates and the use of linear *RPck* rows rather than polar active coordinates. Ferrocene protects center-based special coordinates and high-symmetry point-group projectors; the η^3 case protects consumption of an explicit multi-atom ligand center. Weak-complex tests protect the fragment-coordinate policy and the analytic **B** rows of interfragment translation and orientation coordinates.

The domain-first extension adds targeted invariants beyond rank closure. The Python and Fortran paths must produce the same ordered primitive chart and Wilson rows for an equivalent XY_3 rotor; the XY_2 control must contain two in-plane rows, one owned out-of-plane row and one normalized twofold rotor. The edge corpus also verifies rejection of cyclic XY_2 centers, heterosubstituted branches with the same element and degree, fully symmetric centers lacking a unique Z , linear anchors and duplicated ownership. Ethane exercises adjacent XY_3 domains, whereas the established fused, bridged, spiro and cage corpus verifies that local-center recognition does not pre-empt ring or butterfly ownership.

Table 7 gives a representative self-consistency slice; the complete validation table is reported in SI Sec. S6. The first three molecular rows shown here include comparisons with a separate Fortran control implementation; the reported residual is the Python–Fortran (PF) Wilson-row-space residual between two internally maintained non-redundant bases that share the same scientific specification but not the same implementation path. The narrow range of Δ_{PF} values is expected: these residuals are dominated by the fixed row-space audit tolerance and floating-point floor of the comparison, not by a molecule-specific physical error. Their role is to show that the two implementation paths saturate the same numerical equivalence criterion across systems of different size and symmetry. The formic-acid–water row uses the `SPECIAL_COORDINATES` fragment path, for which the covalent Fortran control reference is not the relevant comparison; there the table reports direct analytic **B**-rank closure and the number of protected special coordinates. Because this weak-complex probe is C_1 , $\epsilon_{\text{SALC}} = 0$ is trivial by construction and should be read only as the absence of nontrivial SALC blocks,

not as a symmetry-adaptation test. The semantic-layer row in the complete SI table is a unit-level contract test of the PROTECT/OBSERVABLE layer introduced in the section on coordinate functions and downstream models; it exercises duplicate rejection, row-equivalent substitution and the fail-fast protected-prefix rule.

The intermolecular projector is tested separately in C_s , C_{2v} , C_{3v} and C_i . In every case the projected analytic **B** reaches the full vibrational rank, retains exactly the family counts of the native non-redundant basis, and passes the Cartesian irrep-covariance audit. The wider symmetry corpus additionally covers D_{2d} , D_{5h} , D_{5d} , T_d , O_h and icosahedral operation classes. These are construction tests, not electronic-structure benchmarks.

Table 7: Representative SONIC validation. Ranks are final/target; Δ_{PF} is the Python–Fortran row-space residual, ϵ_{SALC} the saved-coefficient norm error and ϵ_{FD} the analytic–finite-difference row error.

System	Stress test	PG	Rank/closure	Numerical check
Azulene	Non-benzenoid fused ring, butterfly	C_{2v}	48/48; 6 <i>RPck</i>	$\Delta_{\text{PF}} = 6.44 \times 10^{-9}$; $\epsilon_{\text{SALC}} = 2.22 \times 10^{-16}$
Norbornane	Bridged saturated ring	C_{2v}	51/51; 3 <i>RPck</i>	$\Delta_{\text{PF}} = 6.22 \times 10^{-9}$; $\epsilon_{\text{SALC}} = 2.22 \times 10^{-16}$
Cubane	High-symmetry cage	O_h	42/42; cyclic bends	$\Delta_{\text{PF}} = 6.41 \times 10^{-9}$; $\epsilon_{\text{SALC}} = 2.22 \times 10^{-16}$
Indane	Fused aromatic/alicyclic rings	C_s	51/51	$\epsilon_{\text{FD}} = 4.77 \times 10^{-10}$; butterfly and <i>RPck</i> rows covered
Saccharin	Planar fused heterocycle, butterfly–puckering phase	C_s	45/45; 5 <i>RPck</i>	$\epsilon_{\text{FD}} = 3.91 \times 10^{-10}$; full post-projector B rank 45
Formic acid–water	Weak-complex fragment model	C_1	18/18; 6 protected	three translations plus three orientations; direct B -rank = 18
Intermolecular metry corpus	sym- Family-quota projector	C_{2v} , C_{3v} , C_i	12/12; 18/18; 18/18	native family counts retained; Cartesian irrep covariance passed
Benzene–water	Aromatic–polar fragment model	C_1	39/39; 6 protected	standalone Cartesian input; ring coordinates coexist with the fragment block
η^3 allyl–PdCl	Supplied η^3 ligand center	C_1	24/24; 1 protected	$q_{\eta^3} = 1.8000 \text{ \AA}$; analytic center–atom row retained

A related case emerged during extended independent use of the code by collaborators. An earlier dihedral-based representation of endocyclic out-of-plane deformations caused the internal-to-Cartesian back-transformation to fail to converge for near-planar fused-ring systems, including an indane/indene-type scaffold with an $\text{sp}^3 \text{CH}_2$ center in the five-membered ring. This behavior is consistent with the collective dihedral redundancy and branch problem

discussed in Sec. 7.5: the ill- conditioned ring chart reflected the coordinate choice, not a failure of SONIC rank or symmetry construction. The independent test therefore motivated the replacement by the curvilinear U out-of-plane source, which retains a finite signed row at planarity. A second regression, saccharin, tests the coupled case in which one fused-ring butterfly and five *RPck* rows all transform as A'' . Phase-covariant projection now preserves all 45 coordinates and gives Cartesian rank 45; the butterfly is the difference, rather than the rank-deficient sum, of its two oriented source rows. The U -based form is the version validated throughout this work, including the indane and saccharin entries of Table 7.

The finite-difference audit in the pagodane and indane rows of the complete SI table is applied after the complete stored SONIC chain rule, rather than separately to selected primitive formulas. It therefore covers local SALCs and the linear combinations defining cyclic, butterfly and puckering coordinates. For all 114 pagodane and all 51 indane coordinates, mathematical display vectors are generated with the Wilson matrix frozen at the reference structure, $\Delta \mathbf{x}_k = \mathbf{B}_0^+ \mathbf{e}_k \Delta q_k$. The two frames $\mathbf{x}_0 \pm \Delta \mathbf{x}_k$ are therefore exactly symmetric and freeze every other SONIC to first order; they are not presented as nonlinear back-transformations. The amplitude is reduced if necessary to preserve perceived connectivity and proper orientation, keep the maximum atomic displacement below 0.10 Å, and change no reference bond by 15% or more. Both accepted frames are checked against the frozen input topology, and the number of complete topology evaluations is retained in the diagnostic record. An optional mass-weighted right inverse, $\mathbf{M}^{-1/2}(\mathbf{B}_0 \mathbf{M}^{-1/2})^+$, provides the corresponding minimum-mass-norm vector without changing this first-order contract. Its mass provenance is explicit: the user can select average atomic masses, default spectroscopic isotopes, masses stored with a Cartesian Hessian, or a named isotopologue; the automatic choice prefers Hessian masses when they are available and otherwise uses average masses. The mathematical step is distinct from the interactive display amplitude: the viewer normalizes each half-vector to a user-selected maximum atomic displacement in Å. The machine record reports row-normalized conditioning, Cartesian amplification, participation ratio, significantly moving atoms and

the dominant atom. It also stores the center, ring or bond domain, local pseudogroup, local irrep and independent local/global total-symmetry flags whenever those labels apply. Ring motions additionally retain the canonical ring order and primitive coefficients, which are highlighted with their displacement arrows in the viewer. The tests consume every generated three-frame XYZ trajectory and vector table through the same data loader used by the GUI. The versioned `matrix.smith.sonic.diagnostics.v2` manifest has a distributed JavaScript Object Notation (JSON) Schema and dependency-free validator; the GUI loader remains compatible with archived version-1 manifests. For large coordinate sets the same viewer can filter by coordinate family, local domain, local pseudogroup and local or global total-symmetry character.

The local-pseudosymmetry validation is summarized in Table 8; the complete 20-model CN 5–9 table, including template scores, margins and thresholds, is given in SI Sec. S3. The first probe is intentionally globally C_1 : it contains three equivalent C–H bonds at one carbon and an asymmetric C–F/C–O substituent framework. The purpose is to verify that a local C_{3v} methyl domain can be used to place its symmetric stretch and bend combinations first without promoting the whole molecule to C_{3v} . A rigid rotation by 0.731 rad about the normalized axis $(1, -2, 0.5)$ leaves the complete center-domain diagnostic records identical. The separate Fortran path independently reports the same C_{3v} center, places A_1 first and closes at 18 coordinates. This test checks the local classification contract; the established tetrahedral control test continues to provide the stricter Python–Fortran ordered-primitive and Wilson-row check. The high-coordination extension is tested on both template geometries for each coordination number from five through nine. Each geometry is repeated after a small angular distortion of one ligand, giving 20 models in each implementation. Python and Fortran retain the same local-group assignment and close at ranks 12, 15, 18, 21 and 24 for CN 5, 6, 7, 8 and 9, respectively. For CN 10 and CN 12, for which no named template is assumed, both implementations select the generic equivalence fallback and close at ranks 27 and 33. Their Cartesian row-space residuals are 1.03×10^{-8} and 1.06×10^{-8} , respectively.

Rotation and ligand permutation leave the accepted-template score and margin unchanged. A separate near-tie stress test, with a deliberately widened absolute acceptance gate, is classified as **AMBIGUOUS** and therefore uses the same generic fallback. Two chemically named controls complement the anonymous polyhedral probes: a slightly distorted SF₆ structure gives the octahedral local block and closes at 15/15, while idealized square-antiprismatic [ZrF₈]⁴⁻ gives *D*_{4d} and closes at 21/21. The former uses explicitly relaxed descriptor and distance thresholds (0.1 and 0.05 Å), illustrating that pseudosymmetry is a declared construction choice rather than an implicit relabelling.

Table 8: Local-pseudosymmetry validation. The point group is exact; the local domain identifies only the tested center or ring.

System	PG	Local domain	Rank	Diagnostic check
<i>C</i> ₁ methyl probe	<i>C</i> ₁	center <i>C</i> _{3v}	18/18	identical after rigid rotation; Python and Fortran put local <i>A</i> ₁ first
Azulene	<i>C</i> _{2v}	two fused-ring domains	48/48	ring domains 1 and 2 recorded; one shared edge assigned once
CN 5–9 centers	<i>C</i> ₁	ten local polyhedral templates	12–24	20 ideal/distorted models; Python–Fortran group and rank agreement
SF ₆ , [ZrF ₈] ⁴⁻	<i>C</i> ₁ , <i>D</i> _{4d}	octahedral and square-antiprismatic domains	15/15, 21/21	frozen template, score, margin and thresholds recorded

Table 9 gives a representative validation slice inspired by examples of the general Gaussian 16 GIC facility. These are not comparisons of optimizer performance; they test whether one particular subset of chemically meaningful variables illustrated in that Supporting Information can also be produced automatically as SONIC contract objects or coordinate functions. The complete motif table and numerical checks are reported in SI Sec. S4. For the ferrocene row, the lower cyclopentadienyl ring is rotated through a five-point torsional scan from the eclipsed *D*_{5h} form to the staggered *D*_{5d} form. The unprojected SONIC contract closes the full rank at every grid point, while the high-symmetry endpoints are covered separately by the projector tests in Table 7. The sparse-**B** density reported there is the fraction

$$n_{\text{nz}}/(n_{\text{GIC}} 3N)$$

of nonzero derivative entries in the stored Wilson matrix; it is a compactness diagnostic for the sparse representation, not an additional rank criterion.

Table 9: Automatic SONIC construction of selected coordinate objects supported by the general Gaussian 16 GIC syntax.

SI motif	SONIC object	Check
Formamide proton transfer	Generated R_{NH} , R_{OH} primitive signatures plus Σr and Δr functions	Function values and sparse chain-rule rows reproduce the sum and difference derivatives.
Ferrocene cyclopentadienyl torsion	Ring-center/Fe special coordinates retained through a 0, 9, 18, 27, 36° ring rotation scan	Each grid point has rank 57/57, analytic B rank 57, stable ring-center identities and sparse B density 0.095–0.101.
Cyclohexane ring flip coordinate	Automatically generated D_{3d} ring-puckering block $A_{2g}RPck001 + E_uRPck001/002$	No hand-written Gcyc , Ccyc or Scn0 input is supplied; the model closes rank 48/48 and exports a Gaussian-readable puckering model.

13 Serialization and External-Consumption Probes

The preceding tests establish rank closure and preservation of symmetry labels. The compact application probes ask a different question: whether an independent program can parse and use coordinates emitted from the same frozen object. They are deliberately small calculations because their role is external consumption, not validation of an optimization or potential-energy-surface algorithm belonging to SMITH.

13.1 Gaussian 16 as an external GIC interpreter

The most direct external test is to let Gaussian 16 use one serialization of the SONIC coordinates as its optimization coordinate system. In this probe SONIC generates the Gaussian-16-readable **ReadAllGIC** block, and Gaussian 16 reads that block, evaluates the GIC derivatives with its own machinery and performs the optimization. Gaussian 16 is used specifically because its GIC language is general enough to express sums, differences and nonlinear coordinate functions and therefore tests SONIC definitions directly rather than through a bespoke adapter. This use of GIC is only a compatibility probe and is not intended to characterize the full range of coordinate definitions supported by Gaussian 16.

No hand-written surrogate coordinates are introduced. The three systems in Table 10 were chosen because they exercise ring and cage coordinates while remaining small enough for repeated external checks. The native SONIC contract includes signed out-of-plane rows and block-local multi- U *RPck* coordinates. Because Gaussian 16 Rev. C.01 cannot represent these objects natively, they are translated only at the terminal serialization boundary; the optimized Cartesian structure is then re-evaluated in the frozen native contract. Exact `ReadAllGIC` route syntax, the U -to-improper convention and the treatment of non-totally symmetric rows are documented in SI Sec. S2. The calculations reported here used the commercial Gaussian 16 Rev. C.01 installation at the Hartree–Fock (HF)/STO-3G level. No feature available only in the development version described in Ref. 11 was used.

The relevant observations are that Gaussian 16 accepts the automatically generated coordinate block and reaches a stationary point in three to nine steps for saturated ring and cage systems with different symmetry and rank.

Table 10: Gaussian `ReadAllGIC` optimizations with SONIC coordinates at HF/STO-3G. The Gaussian-rank column is blank because these logs do not report the $NTRed/NRank$ diagnostic.

System	PG	SONIC rank	Gaussian $NTRed/NRank$	Steps	E_{final} / E_h
Cyclohexane	D_{3d}	48/48	—	5	−231.482671171
Norbornane	C_{2v}	51/51	—	9	−268.893823819
Cubane	O_h	42/42	—	3	−303.781399977

A separate comparison using `CalcFC`, reported below and reproduced in the Supporting Information for norbornane, is distinct from this no-`CalcFC` solver-consumption table.

A symmetry-enabled norbornane variant contains the complete rank-closing block but activates only its 15 A_1 rows. Gaussian 16 accepts this serialization and reaches a C_{2v} -preserving stationary point in eight steps; the full input and output excerpts are given in SI Secs. S1–S2.

The same export path was then compared directly with three standard Gaussian optimization choices. Two very small systems verify that no special handling is needed when the

coordinate problem is simple; norbornane and camphor add larger bridged-ring cases, with camphor reaching 75 vibrational degrees of freedom. These calculations are not a benchmark of electronic structure theory or a claim that one coordinate choice always minimizes the iteration count. The SONIC Gaussian input uses an initial force-constant evaluation, as is common for difficult non-redundant coordinate optimizations. They test the narrower point that SMITH can generate a Gaussian-16-readable SONIC coordinate model without manual GIC transcription and that Gaussian 16 can optimize from it using the `ReadAllGIC` parser. At HF/STO-3G, water and hydrogen peroxide give the same or nearly the same iteration counts across the tested coordinate choices. Norbornane converges in four steps with Gaussian redundant internals and Cartesians, six steps with a sequential Z matrix and two steps with the automatically generated SONIC block. Camphor converges in eight steps with Gaussian redundant internals, 14 Cartesian steps, four SONIC compatibility steps and 33 steps for a sequential Z matrix. Table 11 gives the numerical values plotted in Fig. 5, so the prose, table and figure refer to the same diagnostic set. The point of this diagnostic is therefore not that SONIC is always the shortest optimizer path in Gaussian 16, but that the automatically generated GIC block is accepted by the Gaussian 16 `ReadAllGIC` parser and remains competitive. The Gaussian 16 run is a parser/optimizer compatibility probe, not a change in the scientific coordinate contract: the non-redundant SONIC rows remain the reference used to interpret and compare output geometries.

Table 11: Gaussian 16 HF/STO-3G optimization steps from Fig. 5: built-in redundant coordinates versus the generated SONIC `ReadAllGIC` block.

System	PG	SONIC rank	Default	Cartesian	Z matrix	SONIC
Water	C_{2v}	3/3	4	4	4	5
Hydrogen peroxide	D_{2h}	6/6	6	6	5	6
Norbornane	C_{2v}	51/51	4	4	6	2
Camphor	C_1	75/75	8	14	33	4

A separate native-coordinate check retains the curvilinear U representation throughout. Camphor contains one native $U(6, 4, 7, 8)$ row and converges in eight HF/STO-3G steps with

the internal test evaluator. The corresponding Gaussian 16 input is necessarily less compact because U and multi- U ring combinations are terminally translated into supported improper and periodic-dihedral objects. This translation does not redefine the scientific contract: U , multi- U $RPck$, $QPck$ and ΦP remain the native coordinates against which the resulting geometry is interpreted. SI Sec. S2 gives the exact atom-order convention and the frozen rows of the 51-coordinate norbornane example.

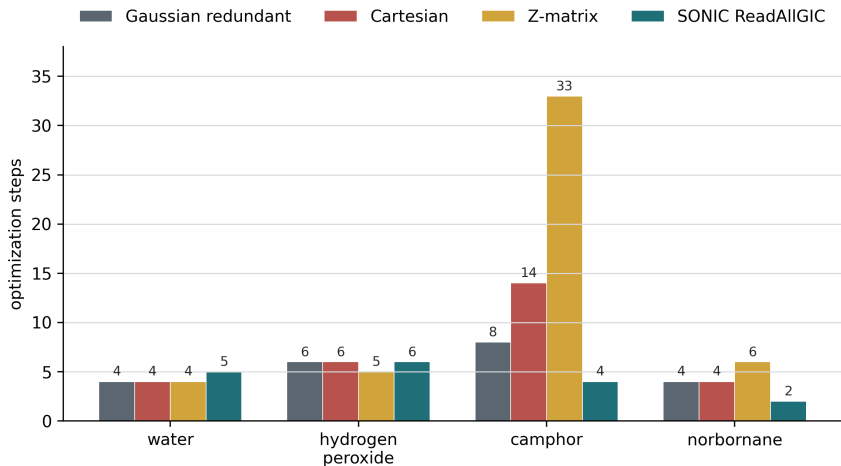


Figure 5: Gaussian 16 HF/STO-3G optimization steps for built-in and generated SONIC ReadAllGIC coordinates. Water and hydrogen peroxide test input portability; norbornane and camphor test bridged-ring cases.

13.2 Ring Puckering Branch

This probe uses the same cyclobutane definition to define a one-dimensional ring-puckering branch. The coordinate is the alternating carbon-ring puckering amplitude associated with the D_{2d} ring source block. Single-point Gaussian 16 HF/STO-3G energies were evaluated along a fixed substituent-frame branch. This is intentionally not a relaxed conformational profile; it is a test that a named puckering coordinate can be used as a reproducible sampling axis while the basis retains the same ring-source identity.

Figure 6 illustrates the result. The minimum on this fixed branch is near $q = 0.04 \text{ \AA}$, matching the optimized reference geometry. Larger puckering displacements raise the energy

smoothly. In a DVR or PES-fitting application, the same base coordinate could be replaced by a Cremer–Pople amplitude or phase function $y = f(\mathbf{q})$, while the analytic chain-rule row would still descend from the frozen SONIC **B**.

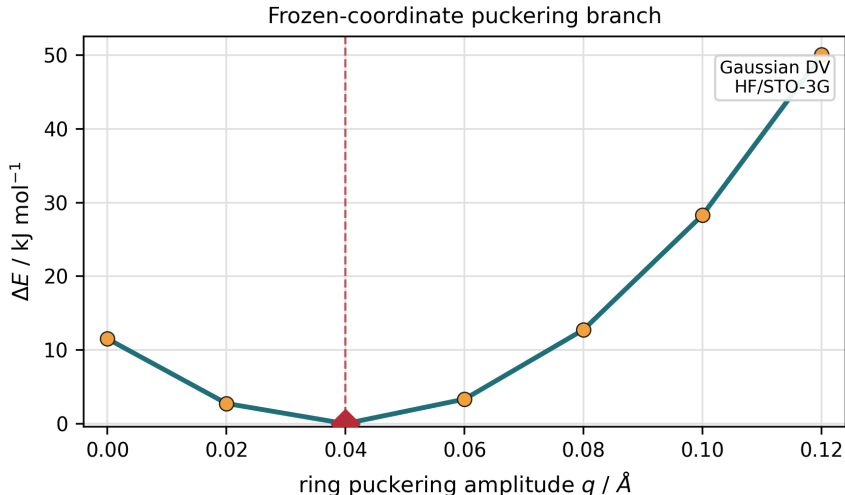


Figure 6: Fixed-frame cyclobutane puckering branch generated from the frozen SONIC ring coordinate and evaluated with Gaussian 16 HF/STO-3G single-point energies. The branch is a coordinate diagnostic, not a relaxed conformational energy profile.

13.3 Portable Chair-to-Boat Puckering Export

The final probe tests a stricter export question, using the same cyclohexane ring-flip motif discussed in the Gaussian GIC paper and its Supporting Information. There, the puckering coordinate is encoded by an explicit molecule-specific input block containing variables such as `Gcyc`, `Ccyc` and `Scn0`. Here no such block is provided. SONIC starts from the molecular graph and geometry, generates the ring-puckering path, and then constructs a fixed-frame cyclohexane puckering path from the chair q_3 component to the boat q_2 component. For each point it generated two independent external representations of the same geometry: a portable Cartesian input, usable by any electronic-structure program, and a Gaussian `ReadAllGIC` input containing the automatically generated coordinate contract. Gaussian B3LYP/6-31+G* single-point energies were then evaluated from both inputs.

All five points have rank 48/48 and contain no out-of-plane GIC terms, so the comparison

uses only syntax supported by standard Gaussian `ReadAllGIC`. Figure 7 shows that the two energy traces are numerically identical: the largest Cartesian–`ReadAllGIC` difference is $3.0 \times 10^{-3} \mu E_h$, about $8 \times 10^{-6} \text{ kJ mol}^{-1}$. The large rise along the path is not a chair–boat barrier; it is the expected cost of an unrelaxed fixed-frame puckering deformation. The result instead shows that the same SONIC-generated large-amplitude coordinate path can be exported either as ordinary Cartesians or as an automatically generated Gaussian GIC model without changing the electronic energy surface being sampled.

Because the Cartesian branch is ordinary molecular geometry, it is not tied to Gaussian. The same three representative points were also evaluated on a local workstation with ORCA and Molpro at the PBE0/def2-SVP density-functional-theory (DFT) level, a different level from the B3LYP/6-31+G* Gaussian check. The absolute energies differ slightly between codes, as expected from independent DFT implementations and defaults, but the relative fixed-frame puckering profile is the same at the level needed for this export test; the complete energies and relative-energy comparison are reported in SI Sec. S5. This verifies the intended division of labor: SONIC constructs the model and coordinates, while external electronic-structure programs consume the resulting Cartesian geometries or GIC serializations. It confirms that the coordinate definition is independent of the electronic-structure implementation.

14 Discussion

SONIC can be viewed as a constructive generalization of natural internal coordinates. Its novelty is not the use of Wilson rows, SVD, Gram–Schmidt or SALC projectors separately, but the deterministic protocol that combines them while respecting locality, symmetry, semantic priority, protected coordinates, ring source spaces and fragment coordinates. The method is deliberately not a single global orthogonalization: local and block-local decisions are followed by a rank audit in the physical Wilson **B**-row space. The structural properties stated above are correspondingly modest: relabelling covariance, determinism on a frozen state,

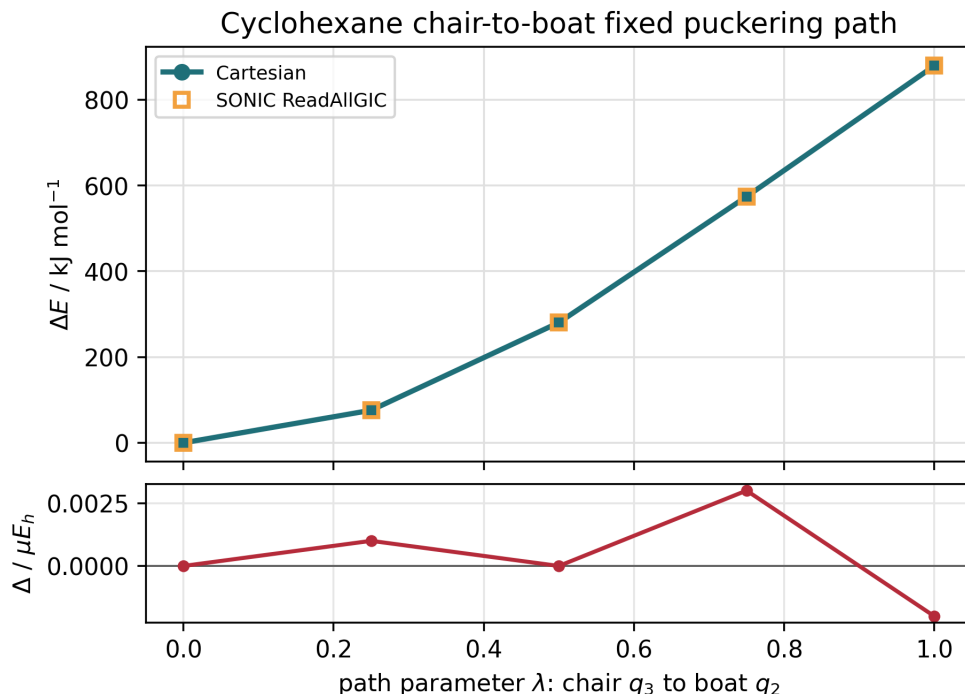


Figure 7: Cyclohexane fixed-frame chair-to-boat export at B3LYP/6-31+G*. The upper panel compares Cartesian and generated SONIC `ReadAllGIC` inputs; the lower panel gives their energy difference.

protected-coordinate preservation when rank allows it, block-local scaling invariance and local stability within a fixed molecular-state stratum. Each follows from the fixed candidate order, normalized Gram–Schmidt rank tests and homogeneous point-group projectors.

The method should therefore be read as a documented coordinate-selection protocol, not as a proof that only one internal-coordinate basis is possible. Normalized Wilson-row testing removes the most direct dependence on raw physical units in the final rank decision, but the protected-first family order remains a chemical modelling choice. The point is that this choice is explicit, reproducible and reported, rather than hidden in primitive weights chosen before a global diagonalization.

This is also the practical answer to the possible “black box” criticism: the output stores primitive definitions, coefficients, family labels, irreducible representations, rank diagnostics and analytic derivative rows. Downstream calculations can also compare the GIC active space with a symmetry-adapted Cartesian nullspace that is not built from GIC primitives;

agreement shows that the chemical conclusion depends on topology, symmetry and rank, not on a hidden generator choice.

The second-order boundary prevents a different category error. The ordinary Wilson matrix describes the tangent map $d\mathbf{q} = \mathbf{B} d\mathbf{x}$; its derivative describes how that tangent map curves as the geometry changes. Equations (20) and (21) explain why a Hessian-transforming application may require $\mathbf{B}'$ away from equilibrium. They do not make $\mathbf{B}'$ a SONIC construction product. At an exactly stationary harmonic point the gradient contraction vanishes, while at a constrained or nonstationary point the consuming program evaluates the required derivative slices from the frozen coordinate formulas. SMITH therefore remains a first-derivative coordinate constructor rather than a general Hessian engine. In MATRIX this second-order consumer is ARCHITECT.

Table 8 adds a complementary check that is not available from molecular point-group tests alone: a chemically useful local totally symmetric coordinate remains identifiable in a C_1 molecule, but it does not acquire a false molecular A_1 label. The ring cases show the same separation between a cyclic local subgroup and the exact molecular group. The local thresholds can change a domain partition near a boundary; because the tolerances, confidence, participating atoms and kept/pruned status are serialized, that change is auditable rather than implicit. The same table also removes the former high-coordination refactor boundary: both implementations now recognize the two documented templates for every coordination number from five to nine, while the generic construction closes CN 10 and CN 12 without introducing additional named polyhedra. The ligand-equivalence fallback is also selected for unrecognized, ambiguous or chemically split environments. Because the score, second-template margin, thresholds and selected status are serialized, this choice is fixed and auditable across independent uses of the same coordinate contract.

The internal validation in Table 7 establishes rank closure, symmetry-projector normalization and Python–Fortran row-space equivalence for representative cases. This is an implementation-level cross-check inside the same development lineage, not a comparison with

an external published natural-coordinate program. Table 9 addresses the more external-facing question inspired by the Gaussian 16 GIC Supporting Information: can a selected set of chemically specialized variables also be generated automatically as SONIC objects? Gaussian 16 supplies a general expression and derivative language supporting many kinds of coordinate definitions; SONIC supplies one optional source of definitions, together with semantic interning, sparse derivative propagation and frozen provenance. In this work, GIC syntax is used as a serialization and solver interface for that particular source. The cyclohexane check makes this distinction concrete: the Gaussian-style puckering model is available to the external solver, but its coordinate definitions are not typed by hand.

14.1 Computational Cost and Remaining Validation

The expensive step is construction. For m primitive candidates, target rank r , N atoms and z nonzero derivative entries, the current dense rank audit costs $O(mr^3N)$ after local reductions have been formed. In ordinary molecular graphs m and r are $O(N)$, giving a conservative cubic build-time bound. This is a pessimistic upper limit: ring, coordination and fragment operations are confined to typed source blocks, and symmetry projection is applied after rank reduction rather than to a full heterogeneous primitive matrix. Once the definition is stored, an application evaluates sparse Wilson rows with cost $O(z)$, plus any chain-rule work it elects to perform. This separation is the reason the code keeps coordinate construction and **B** evaluation as distinct services. Table 12 times only these SMITH operations; it contains no **B'**, Hessian transformation or internal-to-Cartesian realization benchmark.

Table 12 and Figure 8 give a first scaling study for the present implementation. The values are median wall times from five Python runs on a Linux workstation (two AMD EPYC 7543 32-core processors, 128 logical central processing units (CPUs) with simultaneous multithreading, and 1.0 TiB random-access memory (RAM)), measured with `tracemalloc`; they are therefore engineering diagnostics rather than hardware-independent constants. The timings were collected as single-process Python construction jobs, not as parallel throughput

benchmarks. The test set contains three slices: a controlled fused polycyclic-aromatic-hydrocarbon (PAH) series from benzene to coronene, a ring/cage topology set, and two metallocene special-coordinate cases. The intended split is visible. Once the coordinate contract exists, one analytic Wilson-**B** evaluation costs milliseconds to a few 10^{-2} s for the tested covalent and ring systems. Construction is larger because it includes candidate generation, protected-first rank reduction, optional symmetry projection and writing the frozen sections.

In the controlled PAH growth series, a log-log fit gives a build-time slope of 1.48 versus atom count ($R^2 = 0.79$) and a **B**-evaluation slope of 0.94 versus the number of sparse nonzero derivative entries ($R^2 = 0.94$). The left panel of Figure 8 reports a different diagnostic fit: build time against primitive-candidate count over the non-special systems. That broader fit has slope close to unity but is intentionally not used as a universal law, because cubane and the high-symmetry PAHs show that topology and point-group projection can dominate a simple one-variable model. Across the non-special systems, peak Python memory scales approximately linearly with the number of stored primitive rows (slope 1.06, $R^2 = 0.81$). The ferrocene entries are the expected expensive cases in this set because the special-coordinate mode retains a large ring-center/metal primitive source space through symmetry projection before reducing to the 57-dimensional active contract.

Table 12: SMITH/SONIC scaling. Times are medians of five Python runs; build includes construction, rank reduction, optional symmetry projection and output, whereas **B** is one analytic evaluation of the stored contract.

Series	System	<i>N</i>	Cand.	Rank	Build / s	B / s	Peak MiB
PAH	Benzene	12	30	30	0.131	0.0054	0.23
PAH	Naphthalene	18	48	48	0.139	0.0093	0.28
PAH	Anthracene	24	66	66	0.243	0.0139	0.51
PAH	Phenanthrene	24	66	66	0.211	0.0131	0.52
PAH	Pyrene	26	78	72	0.256	0.0162	0.59
PAH	Coronene	36	120	102	0.722	0.0343	1.08
Ring/cage	Cubane	16	54	42	0.400	0.0139	0.45
Ring/cage	Norbornane	19	52	51	0.099	0.0120	0.29
Ring/cage	Cyclooctane	24	66	66	0.142	0.0201	0.53
Ring/cage	Camphor	27	76	75	0.145	0.0151	0.44
Ring/cage	Spiro	31	87	87	0.253	0.0225	0.75
Special	Ferrocene D_{5h}	21	651	57	3.324	0.0642	2.83
Special	Ferrocene D_{5d}	21	656	57	3.182	0.0653	2.59

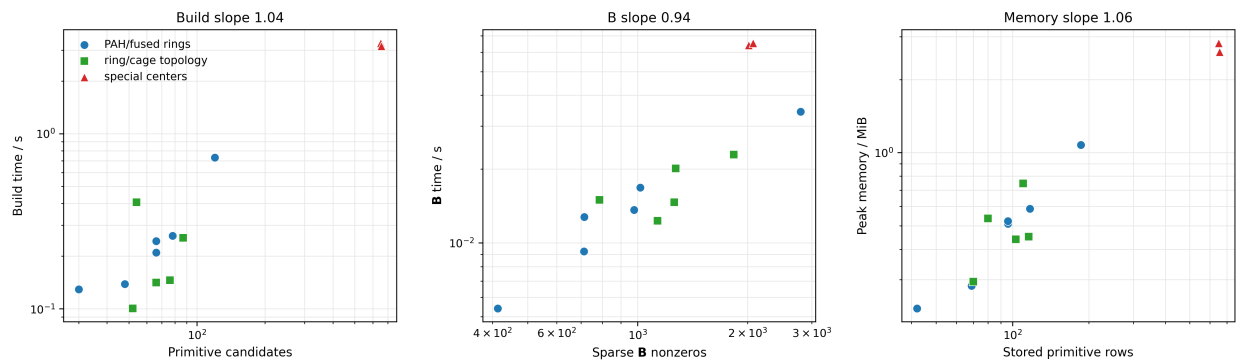


Figure 8: Scaling diagnostics from Table 12: build time, analytic **B**-evaluation time and peak traced memory. Ferrocene is separated because its semantic source space greatly exceeds its active rank.

The study is still finite and implementation-specific. It samples one PAH growth series plus chemically diverse topology and special-coordinate probes; peak memory is Python allocator memory observed during one build-plus-**B** trial. A release benchmark should add larger controlled series for fragment count and for user-protected semantic coordinates, but the present data already quantify the construction/use separation on the molecular classes exercised in the paper.

Other validation items remain open. The local-stability statement gives an existence threshold δ in terms of decision margins and Lipschitz constants, but no general numerical values are reported for a real molecule. The local-SALC probe now supplies one rigid-rotation invariant and fixed-threshold classification test, but it is not yet a statistical tolerance-sensitivity study over thermally distorted structures. Near-threshold point-group perception has the same limitation: the implementation records operation residuals, minimum margins and warning flags, but quantitative tolerance-sensitivity tables remain future work. Likewise, the Gaussian tests are export and solver-consumption tests, not a comparison with an external natural-coordinate generator. The semantic-layer unit probes cover protected-prefix rejection; a chemically realistic overconstrained example would still be useful.

The main limitation is no longer the absence of pseudo-bonds in the code path, but the current validated scope of the pseudo-cycle tests. Protected fragment and special-center

coordinates are already suitable for weak complexes and coordination motifs, while the pseudo-bond baseline covers typed hydrogen-bond closures and bifurcated contacts. What remains to be expanded is the corpus of symmetry-rich intermolecular cycles and the numerical stress testing of those pseudo-cycle rows under geometry changes.

14.2 Scope and Limitations

SONIC is intended for high-accuracy quantum-chemical, spectroscopic and refinement workflows in which the coordinate model carries chemical information and must be reused across modules. It is not designed as a large-scale thousand-atom molecular-dynamics coordinate generator, a generic classical force-field engine or a machine-learning featurizer. Users who need a frozen, auditable, symmetry-aware coordinate object for molecules, rings, fragments or coordination motifs are the target audience; users who only need a black-box optimizer coordinate for very large systems should normally use a lighter optimizer-specific representation.

The construction is conditional on a fixed molecular-state stratum. Bond breaking or forming, proton transfer that changes the covalent graph, and coordination rearrangements require separate contracts or a future state-interpolation model. An empirical-valence-bond (EVB)-like or subspace-interpolation layer may be possible, but it is not part of the present theory.

The coordinate definitions remain frozen, whereas their Wilson rows are evaluated at the current geometry. For fragment rotations, local-chart rebasing also transports values and rows covariantly. Large transformations that change topology, fragment membership or symmetry stratum still require a new contract; adaptive interpolation between such discrete molecular states is outside the present scope.

The same locality qualification explains why $\mathbf{B}'$ is outside the SMITH contract. A second-order consumer can obtain it analytically from the stored primitive formulas and SONIC coefficients, but that consumer must define its own treatment of singular or unsupported charts and report any numerical fallback. Near linear-angle singularities, fragment-rotation

chart boundaries or a topology change, a divergent $\mathbf{B}'$ is coordinate curvature, not a physical force constant and not a failure of SONIC construction.

Standalone operation has a related, explicit scope boundary. Supplied topology or primitive sections are authoritative and preserved, while the bundled Cartesian frontend uses a revision-pinned ORACLE subset to construct ordinary topology, point-group operations, atom permutations and PICs. Results that depend on advanced continuous atom equivalence, nondefault quasi-symmetry decisions or special interaction centers require those data in the input state. This is a capability distinction between input profiles, not a runtime dependency of SMITH on a separate application. It also means that changes to the embedded perception subset require an explicit standalone version update and renewed equivalence tests against ORACLE.

The priority-first rule is a chemically motivated modelling principle, not a mathematical uniqueness result over all possible coordinate bases. Alternative priority orders would define different, still reproducible contracts. The claim made here is that the priority order is explicit, versioned and diagnosed, and that special coordinates are not silently displaced by ordinary coordinates when they are compatible with the target rank.

Collective XY_3/XY_2 rotors are a conservative optional construction, not a claim that every threefold- or twofold-looking geometry has that local symmetry. Recognition uses the frozen graph environment and exact terminal permutability; geometry supplies admissibility and nonlinearity tests but does not by itself establish branch equivalence. When the domain is ambiguous, heterosubstituted, cyclic in the XY_2 case, or lacks an anchor, the contract records the fallback and retains one ordinary dihedral.

Finally, determinism is conditional on the frozen molecular state: geometry, topology, symmetry operations, continuous atom perception, synthon descriptors, tolerances, semantic grammar version and contract schema version. A different perception model or tolerance policy may produce a different contract; the scientific requirement is that such a change is visible in the provenance rather than hidden in an implementation choice.

15 Conclusions

Starting from a frozen topology and redundant primitive/**B** contract, SONIC provides a general, frozen and symmetry-aware construction of non-redundant internal coordinates. Its main contribution is a reproducible coordinate object whose ownership is decided before numerical rank completion. Stretches establish the frame; rings, XY_3/XY_2 centers, fragments and interaction centers build complete natural blocks; only the remaining primitives generate the ordinary chart. This domain-first order integrates natural-coordinate locality, analytic Wilson-**B** rank control, local A_1 -first pseudosymmetry blocks and block-local point-group symmetrization without allowing a chemically named motion to be replaced by an opaque row-space equivalent. The normalized collective rotors further show how a broad GIC language can be converted into a persistent, non-redundant molecular coordinate model rather than a molecule-specific list of expressions. This is the coordinate object that SMITH reports in human-readable form and can serialize for external use. Vibrational analysis, optimization, sampling, force-field construction and refinement are possible consumers, but their algorithms lie beyond this work. Nonlinear coordinate functions such as inverse or exponential distances, trigonometric angular terms and Cremer–Pople descriptors can be built on top of this base by analytic chain-rule differentiation.

Molecular symmetry remains an input-state responsibility. In MATRIX that state is produced once by ORACLE and consumed unchanged by SMITH. The standalone package achieves independence by embedding a revision-pinned subset of the same ORACLE topology, symmetry and ordinary-PIC kernel. A plain XYZ can therefore produce symmetry-adapted SONIC coordinates without a second installation, while a supplied frozen state remains authoritative. This implementation duplication is deliberately versioned and tested so that it cannot become an undocumented alternative definition of symmetry.

The present construction supports the intramolecular machinery needed for acyclic, cyclic, fused, bridged, spiro and high-symmetry molecules, together with special-coordinate and pseudo-bond models for weak complexes within the validated baseline described above. The

next methodological step is to enlarge the pseudo-cycle validation corpus for intermolecular systems, using the same rank, symmetry and analytic-**B** invariants used by the rest of SONIC. The local-SALC validation additionally shows that center and ring pseudosymmetry can order coordinates without changing the exact molecular point group. The Gaussian 16 `ReadAllGIC` probes show that the present contract is usable by an independent, general symbolic-coordinate evaluator. Gaussian 16 is chosen for this direct test because its GIC syntax is unusually general. SONIC definitions exercise only one part of that general syntax; SMITH neither depends on Gaussian 16 nor modifies, restricts or seeks to replace its internal-coordinate facilities. The puckering and portable chair-to-boat export probes illustrate why the same stored definition is relevant to large-amplitude work within the scope defined above. Full application benchmarks remain outside the present construction paper. The cyclohexane export check further shows that a SONIC-generated puckering path can be represented either as ordinary Cartesians or as Gaussian 16 `ReadAllGIC` input with numerically identical energies. The frozen SONIC contract should therefore be viewed not only as a coordinate representation, but as a reusable molecular representation on which future force-field, multilevel and potential-energy-surface methodologies can naturally be built.

Reproducibility

All reported construction tests use fixed inputs and no stochastic seeds. The immutable standalone snapshot is tagged `v0.1.0rc7`; methods and basis sets are stated with each external probe. Complete commands, dependency versions, revision identifiers, checksums and regeneration instructions are given in SI Sec. S7. The Python implementation is checked against a separate Fortran control path wherever that comparison is applicable.

Data and Code Availability

The manuscript repository at <https://github.com/yogibubu/Smith> contains the scripts, JSON data files, figures and Supporting Information used to generate the tables and application plots, including `data/local_salc_validation.json` and its generating script. The same repository contains a reduced standalone Python package, installation instructions and water, norbornane, formic-acid–water, water-dimer, benzene–water and η^3 allyl–palladium examples. Its `smith-sonic build` command accepts a complete validated `xyzin` state, an extended XYZ carrying `TOPOLOGY` or `PRIMITIVES`, or a plain Cartesian input through the bundled perception frontend. The provider-neutral `--require-frozen-state` option requires the complete contract. The standalone distribution includes the revision-pinned topology, point-group symmetry and ordinary-primitive frontend and optional Gaussian `ReadAllGIC` export; installing or running another application is not required. The reproducible standalone environment is tagged `v0.1.0rc7`. Repository revisions, the dependency lock and `#SMITH_PROVENANCE` record the code and input profile.

Supporting Information

Standalone input/output reference and executable examples; Gaussian 16 serialization details; extended local-pseudosymmetry and Gaussian-GIC motif validation; ORCA/Molpro cross-code verification; pagodane, indane and saccharin ring-coordinate fixtures; software environments, checksums and complete reproduction commands (portable document format, PDF).

Author Contributions

Vincenzo Barone conceived the method, designed the SONIC coordinate contract, implemented the active code paths used here, performed the calculations and wrote the manuscript.

External refinement and PES-fitting consumers are discussed only as downstream uses of the coordinate definition.

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Notes

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TOC Graphic

